# Agentic AI Regulations Corpus
# **and Implementation Evidence**

Implementation Specification

Ref: `REG-GOV-2026-v1.2`

SAP-OSS AI Governance Team

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# Document status and purpose

This specification formalises the regulations, frameworks, and conformance tooling relevant to agentic AI deployments in the SAP OSS finance stack. Its three goals are:

1. **Make the regulations machine-readable.** Every source document in source/ is accompanied by structured JSON records under `docs/regulations/machine-readable/` that capture sections, normative requirements, and observational claims.
2. **Bind each requirement to evidence in the repository.** The requirement schema carries an `implementation[]` array of file paths, configuration fields, and documentation anchors, plus a `verification[]` array pointing to AI Verify plugins, Moonshot benchmark IDs, or repository tests. A local validator (`docs/schema/validate.py`) fails when any of these paths no longer resolve.
3. **Give reviewers a single printable reference.** This document cross-links the corpus records, the two vendored conformance frameworks, and the code they exercise, so that governance reviews, internal audit, and external assessors can read one PDF instead of navigating three directory trees.

**In scope.** The Infocomm Media Development Authority (IMDA) / AI Verify Foundation *Model AI Governance Framework for Agentic AI* v1.0 (22 Jan 2026); the multi-institution *2025 AI Agent Index*; the Ju & Aral field experiment *Collaborating with AI Agents* (arXiv 2503.18238v3, Feb 2026); and the two vendored open-source conformance tools aiverify (v2.0) and moonshot-cicd (v1.1.0).

**Out of scope.** Binding legislation (EU AI Act, US executive orders); sector-specific prudential rules (Basel, Solvency); deterministic non-LLM agent platforms. Those are tracked separately and may be added as additional regulation records without changing the schema.

**Audience.** AI governance leads, internal audit, model risk, and technical reviewers responsible for evidencing conformance for agentic deployments that use the SAP AI Core / PAL / BTP stack captured in ../tb/ and ../arabic/.

**Relationship to other corpora.** This corpus is a peer of ../arabic/ (Arabic invoice processing) and ../tb/ (trial balance review). Where requirements apply to those workloads, the `implementation[]` field points back to their structured records or spec chapters.

This is a living document. Re-running python3 docs/schema/validate.py –registry-check checks schema conformance and verifies that every code path, plugin, and test id referenced from the requirement records still exists.

# Chapter 1

# Overview and corpus layout

## 1.1 Why this corpus exists

Agentic AI systems — LLM-driven agents that plan across multiple steps, call tools, and act on external systems — are governed by a rapidly evolving mix of regulatory frameworks, empirical research, and conformance tooling. None of these sources alone tells an organisation whether a specific deployment is defensible. The corpus brings the three source classes together, adds a machine-checkable link to the code implementing each requirement, and exposes the result as both structured JSON and this human-readable PDF.

Three concrete failure modes motivated the corpus:

- **Requirements drift.** Regulatory text is updated faster than organisational policies. A machine-readable copy with a hash lets governance teams detect a published update and re-run the downstream validators.
- **Evidence lives everywhere.** A single requirement such as *"bound the agent's tool access"* touches an MCP server, a data-product registry, a CI job, a runbook, and a review checklist. Without a central mapping, audit evidence walks are painful and inconsistent.
- **Conformance tools don't speak the same vocabulary.** AI Verify uses plugin IDs, Moonshot uses benchmark IDs, internal controls use EUC reference numbers, and product teams use file paths. The requirement schema defined here is the translation table.

## 1.2 Three source classes

**Framework — IMDA MGF for Agentic AI.** A 29-page recommended framework organised into four pillars: assess and bound risks upfront; make humans meaningfully accountable; implement technical controls and processes; enable end-user responsibility. It is the highest-normativity document in the corpus.

**Index — 2025 AI Agent Index.** A descriptive inventory of 30 deployed agentic systems scored on legal, technical, autonomy, ecosystem, evaluation, and safety documentation. It supplies transparency requirements rather than normative obligations.

**Research — Ju & Aral (2026).** A 2,234-participant field experiment that produced empirical claims about delegation rates, task orientation, and diversity collapse. We treat these as observational requirements that inform which metrics a deployment should measure.

## 1.3 Two conformance tools

**aiverify (v2.0).** A plugin-based testing framework. Eleven stock plugins implement fairness, robustness, explainability, financial-sector (Veritas), reporting, and process-checklist algorithms. Plugins are referenced from requirement records via the `verification[].plugin` field.

**moonshot-cicd (v1.1.0).** A CI/CD-focused safety evaluation tool exposing four benchmark categories — hallucination, undesirable content, data disclosure, adversarial — behind a single `moonshot run` command. Referenced via `verification[].testId`.

## 1.4 Corpus layout

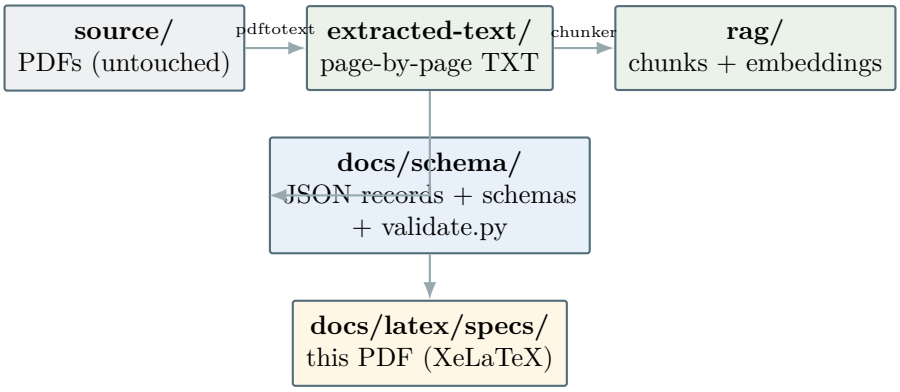


Figure 1.1: Pipeline from source PDFs through to the printable specification.

## 1.5 End-to-end traceability path

Figure 1.2 shows the five-step chain that the validator enforces: from the hashed source PDF, through the extracted text, a structured regulation record, a normative requirement record with cross-references, down to the code or test actually implementing it.

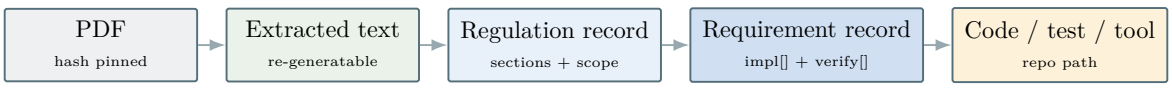


Figure 1.2: End-to-end traceability chain verified by `docs/schema/validate.py`.

## 1.6 What the validator enforces

1. **Schema conformance.** Every record under `docs/schema/regulations/` (`regulation.schema.json`, `requirement.schema.json`, `conformance-tool.schema.json`, `corpus.schema.json`) validates against its Draft-07 JSON Schema using an offline `referencing` registry.
2. **Corpus integrity.** Every entry in `docs/schema/registry.json` resolves to a schema or record on disk, and every record on disk is listed in the registry.
3. **Implementation link existence.** For each `implementation[].path`, the validator resolves the path against the repository root and fails if it is missing (unless the entry is explicitly recorded as `kind: "none"` marking a known gap).
4. **Verification link coherence.** Each `verification[].plugin` must be declared in the aiverify record; each `verification[].testId` in the moonshot-cicd record; each `verification[].path` referencing a test must exist on disk.

The validator is the contract: CI, internal audit, and external reviewers run the same script and see the same results.

| Directory | Purpose |
|---|---|
| `docs/regulations/` (source PDFs) | Binary source documents. Never edited; content hashes tracked in `docs/schema/registry.json`. |
| `docs/regulations/machine-readable/` | Per-source extracted text, RAG chunks, and embedding records (re-upsertable into Qdrant via `machine-readable/scripts/upsert\_qdrant.py`). |
| `docs/regulations/machine-readable/specification-addendum.yaml` | Requirement records (20 requirements) with `implementation[]` and `verification[]` pointers. |
| `docs/schema/regulations/` | JSON Schemas (Draft-07): regulation, requirement, conformance-tool, corpus. |
| `docs/schema/registry.json`, `docs/schema/validate.py` | Master schema index + unified offline validator. |
| `docs/latex/specs/regulations/` | This human-readable PDF (XeLaTeX). |
| `docs/regulations/aiverify-main/` | Vendored AI Verify 2.0 monorepo (Apache-2.0). |
| `docs/regulations/moonshot-cicd-main/` | Vendored Moonshot 1.1.0 monorepo (Apache-2.0). |

Table 1.1: Top-level directory layout.

# Chapter 2

# Canonical data schema

The structured layer uses four Draft-07 JSON Schemas and a central `docs/schema/common/enums.yaml` for controlled vocabulary.

## 2.1 Schema inventory

| Schema | Describes |
|---|---|
| regulation.schema.json | One published framework, standard, guideline, research paper, or index. Captures publisher, version, scope, section tree, and a hash of the source PDF. |
| requirement.schema.json | One atomic requirement or observational claim. Carries `implementation[]` and `verification[]` arrays that bind it to the repository and to the conformance tools. |
| conformance-tool.schema.json | One vendored testing framework. Enumerates plugins, benchmarks, and integration points. |
| corpus-index.schema.json | The top-level index that lists every item in the corpus, typed as regulation, requirement, conformance_tool, or rag_artifact. |

Table 2.1: Schemas under `docs/schema/regulations/`.

## 2.2 Requirement record — the linkage schema

The requirement schema is the key extension that separates this corpus from the earlier finance corpora. Each requirement carries:

- `source`: regulation id, section, page, and (optionally) a verbatim quote.
- `normativeText`: a paraphrased imperative.
- `category` and `obligation`: structured priority (*must*, *should*, *may*, *descriptive*).
- `lifecyclePhase`: where in the agent lifecycle the requirement applies (planning, design, development, pre-deployment, deployment, post-deployment, decommission).
- `implementation[]`: an array of pointers to code, config, documentation, schema, or dataset evidence. A `kind: "none"` entry explicitly records a known gap.

- `verification[]`: an array of pointers to AI Verify plugins (by `plugin`), Moonshot benchmarks (by `testId`), repository tests (by `path`), manual-review entries, or process-checklist items.
- `status`: *implemented*, *partial*, *gap*, *planned*, or *not_applicable*.

Listing 2.1: Example requirement with implementation and verification arrays (excerpt).

```
{
  "requirementId": "REG-MGF-2.1.2-001",
  "source": {
    "regulationId": "imda-mgf-agentic-ai-v1",
    "section": "2.1.2",
    "page": 11,
    "quote": "Bound risks through design by defining agents' limits ..."
  },
  "normativeText": "The agent must operate against an explicit allow-list
                    of tools and external systems; broader access must be
                    denied by default.",
  "category": "bound_risks",
  "obligation": "must",
  "lifecyclePhase": "design",
  "implementation": [
    {"kind": "code",
     "path": "src/intelligence/ai-core-pal/mcp_server/btp_pal_mcp_server.py",
     "symbol": "btp_pal_mcp_server"},
    {"kind": "config",
     "path": "src/intelligence/ai-core-pal/data_products/registry.yaml"}
  ],
  "verification": [
    {"kind": "test",
     "path": "src/intelligence/ai-core-pal/tests/test_mcp_server.py"},
    {"kind": "moonshot-test", "testId": "adversarial_prompts"}
  ],
  "status": "partial",
  "priority": "critical"
}
```

## 2.3 Conformance-tool record

The conformance-tool schema captures the tools themselves so that requirement records can reference them by stable identifiers. Every `plugin.pluginId` and `benchmark.benchmarkId` is part of the vocabulary that requirement records can cite.

Listing 2.2: Excerpt of `docs/regulations/machine-readable/specification-addendum.yaml` (`conformance-tools.aiverify`).

```
{
  "toolId": "aiverify",
  "vendoredPath": "docs/regulations/aiverify-main",
```

```
  "kind": "testing-framework",
  "plugins": [
    { "pluginId": "aiverify.stock.process-checklist",
      "path": "docs/regulations/aiverify-main/stock-plugins/...",
      "purpose": "11-principle process-checklist questionnaire ...",
      "algorithmType": "process" },
    { "pluginId": "aiverify.stock.robustness-toolbox",
      "algorithmType": "robustness" }
    // ... nine more plugins
  ]
}
```

## 2.4 Enum vocabularies

`docs/schema/common/enums.yaml` centralises the controlled vocabularies that recur across schemas. The authoritative values live in the schemas themselves; `enums.yaml` is the human index.

| Enum | Values |
|---|---|
| Regulation type | framework, standard, guideline, statute, research, index, toolkit |
| Normativity | mandatory, recommended, advisory, descriptive |
| Obligation verb | must, should, may, descriptive |
| Requirement category | bound_risks, human_accountability, technical_controls, end_user_responsibility, transparency, evaluation, observation |
| Lifecycle phase | planning, design, development, pre_deployment, deployment, post_deployment, decommission |
| Implementation kind | code, config, doc, schema, dataset, none |
| Verification kind | aiverify-plugin, moonshot-test, test, manual-review, process-checklist, external-audit |
| Status | implemented, partial, gap, planned, not_applicable |

Table 2.2: Controlled vocabularies.

# Chapter 3

# IMDA Model AI Governance Framework for Agentic AI

## 3.1 Source document

**Title.** Model AI Governance Framework for Agentic AI.

**Publisher.** Infocomm Media Development Authority (IMDA) and the AI Verify Foundation, Singapore.

**Version.** 1.0, published 22 January 2026.

**Pages.** 29.

**Source.** source/mgf-for-agentic-ai.pdf (sha256 `b84c8c94…aeef7b`).

## 3.2 Framework structure

The MGF is organised around four pillars:

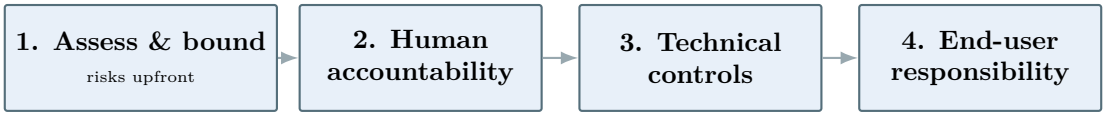


Figure 3.1: The four pillars of the IMDA MGF for Agentic AI.

Each pillar expands into two or three sub-sections (§2.1.1–§2.4.3); the full section tree is captured in `docs/regulations/machine-readable/specification-addendum.yaml` (`regulations.imda-mgf-agentic-ai-v1`) under the `sections[]` array, with page references.

## 3.3 Pillar 1 — Assess and bound the risks upfront

### Summary

Understand the risks posed by agent actions (scope, reversibility, autonomy), determine whether the use case is suitable at all, and — if it is — bound the agent's capabilities by design.

### Requirements extracted

| Req. id | Section | Obligation | Normative text (paraphrased) |
|---|---|---|---|
| REG-MGF-2.1.1-001 | 2.1.1 | should | Assess action scope, reversibility, autonomy per use case. |
| REG-MGF-2.1.2-001 | 2.1.2 | must | Operate against an explicit tool allow-list; deny-by-default. |
| REG-MGF-2.1.2-002 | 2.1.2 | must | Every agent action must be attributable to an identity. |
| REG-MGF-2.1.2-003 | 2.1.2 | should | Set impact-limiting boundaries (read-only default, environment segregation). |

Table 3.1: Pillar 1 requirements.

### How the repository implements them

The tool-allow-list requirement (REG-MGF-2.1.2-001) is implemented by ../../src/intelligence/ai-core-pal/mcp_server/btp_pal_mcp_server.py which enumerates the tools the agent may invoke, and the registered data products in ../../src/intelligence/ai-core-pal/data_products/registry.yaml. Verification leans on the existing unit test at ../../src/intelligence/ai-core-pal/tests/test_mcp_server.py plus a Moonshot adversarial-prompt benchmark.

Identity attribution (REG-MGF-2.1.2-002) currently relies on BTP service keys; this is partial rather than implemented because there is no per-request audit trail that ties every tool invocation back to a named human operator across the full deployment chain.

## 3.4 Pillar 2 — Make humans meaningfully accountable

### Summary

Assign responsibilities by role, adapt human-in-the-loop to address automation bias, name the checkpoints where a human must approve an irreversible action, and audit oversight effectiveness over time.

The TB review corpus is the canonical worked example: the five decision points DP-001..DP-005 (see ../tb/structured/decision-points/) are exactly the human-approval checkpoints required by REG-MGF-2.2.2-001. Their owners and timing are declared in the workflow record ../tb/structured/workflows/tb-review-glo.json.

| Req. id | Section | Obligation | Normative text (paraphrased) |
|---|---|---|---|
| REG-MGF-2.2.1-001 | 2.2.1 | must | Assign lifecycle responsibilities to named roles in a control register. |
| REG-MGF-2.2.1-002 | 2.2.1 | should | Stand up a governance forum with authority to adjust permissions. |
| REG-MGF-2.2.2-001 | 2.2.2 | must | Name each human-approval checkpoint and cite its risk category. |
| REG-MGF-2.2.2-002 | 2.2.2 | should | Measure oversight effectiveness (override rates, review duration). |

Table 3.2: Pillar 2 requirements.

The governance-forum requirement (REG-MGF-2.2.1-002) is closed for the Finance scope by the GCFO Finance AI Council charter (v1.0, ratified 2026-03-15). **Note:** The charter document is maintained in the internal governance repository and is not included in this corpus for confidentiality; reference is by citation only. The GCFO Finance AI Council (GFAIC) is a *federated* body: it operates under Delegation No. 2026-04 from the enterprise-wide Group AI Safety Council (GAISC) and holds binding authority *only* over Finance-scope agentic AI — today the trial-balance review workflow and the Arabic invoice / AP pipeline. Its six voting members are Finance- function delegates (Group CFO chair, Head of Finance Risk, Finance liaisons to CISO and General Counsel, Head of Finance Internal Audit, Finance AI technical lead); the parent body's officers (CRO, CISO, GC, Head of Internal Audit) sit on GAISC, not here, avoiding double staffing. Any incident with cross-function implications escalates to GAISC within two business days. Peer federated councils (People, Customer) exist for other functions and receive GFAIC findings via GAISC rather than directly.

## 3.5 Pillar 3 — Implement technical controls and processes

### Summary

Technical controls across the lifecycle: during development, use controls for planning, tools, and agent protocols; before deployment, test for execution accuracy, policy adherence, and tool use; during and after deployment, continuously monitor.

The trajectory-level evaluator (REG-MGF-2.3.2-002) is implemented at ../../src/intelligence/ai-core-pal/evaluation/trajectory_evaluator.py. It scores each captured agent trace on three independent axes — tool-call correctness, policy adherence, and goal progression — returning a composite `TrajectoryScore` that the Governance Council's incident pack consumes directly. Unit coverage lives at ../../src/intelligence/ai-core-pal/tests/test_trajectory_evaluator.py (13 test cases).

Continuous drift monitoring (REG-MGF-2.3.3-002) is implemented at ../../src/intelligence/ai-core-pal/monitoring/drift_monitor.py. It watches three signals against a frozen pilot-exit baseline — override rate, median review duration, and hallucination rate — and emits

| Req. id | Section | Obligation | Normative text (paraphrased) |
|---|---|---|---|
| REG-MGF-2.3.1-001 | 2.3.1 | must | Technical controls on planning, tools, and protocols. |
| REG-MGF-2.3.2-001 | 2.3.2 | must | Pre-deployment: execution accuracy, policy adherence, tool use, refusals. |
| REG-MGF-2.3.2-002 | 2.3.2 | should | Adopt trajectory-level agent evaluations. |
| REG-MGF-2.3.3-001 | 2.3.3 | should | Staged rollout with canary metrics and rollback path. |
| REG-MGF-2.3.3-002 | 2.3.3 | must | Continuous monitoring with policy-violation / drift alerting. |

Table 3.3: Pillar 3 requirements.

`ok`/`warn`/`critical` levels. The override-rate signal is deliberately two-sided: a *fall* below baseline is the automation-bias signal surfaced by Ju & Aral (2026), a *rise* is model regression. Unit coverage at ../../src/intelligence/ai-core-pal/tests/test_drift_monitor.py (13 test cases).

## 3.6 Pillar 4 — Enable end-user responsibility

### Summary

End users must be informed of the agent's scope and their own obligations, and they need training to preserve tradecraft and mitigate automation bias.

| Req. id | Section | Obligation | Normative text (paraphrased) |
|---|---|---|---|
| REG-MGF-2.4.2-001 | 2.4.2 | should | Disclose permitted actions, data access, user obligations. |
| REG-MGF-2.4.3-001 | 2.4.3 | should | Training covering automation-bias and tradecraft preservation. |

Table 3.4: Pillar 4 requirements.

The training curriculum (REG-MGF-2.4.3-001) is authored at ../training/agent-oversight-curriculum.md: a three-level curriculum (Awareness, Reviewer skills, Oversight depth) keyed to five learner personas, with an explicit reviewer self-check card targeted at the automation-bias failure mode surfaced by Ju & Aral (2026).

## 3.7 Coverage heat map

All four previously-open gaps — the enterprise governance forum (REG-MGF-2.2.1-002), trajectory-level evaluation (REG-MGF-2.3.2-002), drift monitoring (REG-MGF-2.3.3-002), and the training curriculum (REG-MGF-2.4.3-001) — now have concrete artifacts linked from their requirement records and are verifiable by the offline validator. The remaining *partial* rows retain ambition beyond the current implementation scope (for example, a per-request identity audit trail spanning the full deployment chain for REG-MGF-2.1.2-002) and

| Pillar | Reqs | Implemented | Partial | Gap | N/A |
|---|---|---|---|---|---|
| 1 — Bound risks | 4 | 0 | 4 | 0 | 0 |
| 2 — Human accountability | 4 | 3 | 1 | 0 | 0 |
| 3 — Technical controls | 5 | 2 | 3 | 0 | 0 |
| 4 — End-user | 2 | 1 | 1 | 0 | 0 |
| **Total** | 15 | 6 | 9 | 0 | 0 |

Table 3.5: Status of the 15 MGF-derived requirements after the April 2026 gap-closure pass.

remain on the Council's standing agenda.

# Chapter 4

# 2025 AI Agent Index

## 4.1 Source document

**Title.** The 2025 AI Agent Index — Documenting Technical and Safety Features of Deployed Agentic AI Systems.

**Authors.** Staufer (Cambridge), Feng (Washington), Wei (Harvard Law), Bailey (Stanford), Duan (Concordia AI), Yang (UPenn), Ozisik (MIT), Casper (MIT, co-senior), Kolt (Hebrew University, co-senior).

**Venue.** `https://aiagentindex.mit.edu/`; CC-BY-4.0.

**Pages.** 39.

**Source.** source/2025-AI-Agent-Index.pdf (sha256 `c67a3046…0e267f`).

## 4.2 What it catalogues

The Index documents 30 state-of-the-art deployed agentic systems across six documentation categories:

1. **Legal.** Licensing, terms of service, responsible-use policies, redistribution rights.
2. **Technical capabilities.** Supported modalities, tool access, planning horizon, memory.
3. **Autonomy and control.** Spectrum from recommender to operator; human-in-the-loop requirements.
4. **Ecosystem interaction.** External systems the agent integrates with (MCP, web browsers, code editors, e-commerce).
5. **Evaluation.** Published benchmarks, capability scores, safety evaluations, red-team assessments.
6. **Safety.** Safety documentation, risk assessments, residual-risk statements.

## 4.3 Headline finding for governance

The Index's top-level empirical finding is directly relevant to our corpus design:

> *"…we find different transparency levels among agent developers and observe that most developers share little information about safety, evaluations, and societal impacts."*

Our response is the transparency requirement set extracted in Table 4.1.

| Req. id | Section | Obligation | Normative text (paraphrased) |
|---|---|---|---|
| REG-AGENTIDX-3.6-001 | 3.6 | should | Publish per-agent: risk assessment summary, evaluation results, known limitations. |
| REG-AGENTIDX-3.3-001 | 3.3 | should | Document agent autonomy level and map each state transition to that level. |

Table 4.1: Requirements derived from the 2025 AI Agent Index.

## 4.4 How we verify the transparency requirement

Publication is evidenced by this very document plus the structured records. In practice:

- Risk assessment: ../tb/structured/requirements/tb-business-requirements.json, `riskAssessment` field.
- Evaluation results: AI Verify `process-checklist` plugin output plus Moonshot benchmark runs (Chapter 6).
- Known limitations: the gap entries in this specification's coverage tables.

This chapter is cited from requirement record `REG-AGENTIDX-3.6-001` as `docs/regulations/spec/chapters/04-agent-index.tex`, closing the loop.

# Chapter 5

# Empirical evidence: Ju & Aral (2026)

## 5.1 Source document

**Title.** Collaborating with AI Agents: A Field Experiment on Teamwork, Productivity, and Performance.

**Authors.** Harang Ju (Johns Hopkins Carey) and Sinan Aral (MIT Sloan).

**Venue.** arXiv:2503.18238v3 [cs.CY], 5 February 2026.

**Pages.** 59.

**Source.** source/2503.18238v3.pdf (sha256 `b680f584…cd7877`).

## 5.2 Study design

The authors built the *Pairit* platform and randomly assigned 2,234 participants to human-human and human-AI teams producing 11,024 ads for a think tank. Outputs were rated by humans and field-tested on X, which garnered roughly five million impressions. The design is — for this corpus — an unusually strong source of evidence about how human-agent collaboration actually affects output quality, homogeneity, and team dynamics.

## 5.3 Findings relevant to governance

1. **Jagged productivity frontier.** Human-AI teams produced 50% more ads per worker and higher *text* quality; human-human teams produced higher *image* quality. Different sub-tasks need different evaluation metrics.
2. **Delegation shifts.** Participants delegated 17% more work to AI agents than to human partners and performed 62% fewer direct text edits when working with AI.
3. **Diversity collapse.** Human-AI outputs were more self-similar. For creative or generative use cases this is a first-order governance risk.

4. **Communication restructure.** Human-AI collaboration was significantly more task-oriented (+25% task messages, -18% interpersonal).
5. **AI-awareness moderation.** Participants who correctly identified they were paired with AI delegated more and were more task-oriented — the effect is moderated by knowing the collaborator is an agent.

## 5.4 Requirements extracted

| Req. id | Section | Obligation | Normative text (paraphrased) |
|---|---|---|---|
| REG-JU-ARAL-3-001 | 3 | descriptive | Evaluate per-task; do not generalise a single productivity metric. |
| REG-JU-ARAL-4-001 | 4 | should | High delegation + low direct-edit rate is a proxy for automation-bias risk; measure it. |
| REG-JU-ARAL-5-001 | 5 | should | Add an output-diversity metric for creative-output use cases. |

Table 5.1: Observational claims treated as requirements.

## 5.5 Implementation in the corpus

- REG-JU-ARAL-3-001 is implemented by this chapter; it reminds reviewers to look at the per-task KPIs defined in ../tb/structured/kpis/tb-kpis.json rather than a single blended productivity number.
- REG-JU-ARAL-4-001 maps to the override-rate and review-duration KPIs in the TB corpus. When override rates fall well below historical baseline and review duration drops in lockstep, that is the signal to re-audit.
- REG-JU-ARAL-5-001 is marked *not_applicable* for the trial-balance workflow (deterministic review, single narrative per legal entity). We record rather than omit it so that the framework surfaces the requirement automatically when it becomes applicable.

  **Activation trigger.** The requirement flips to `applicable` when any one of the following is true: (a) a deployed agent produces *multi-variant creative output* (e.g., drafting options, marketing copy, generative translation) rather than deterministic reports; (b) the Simula training corpus is used to fine-tune a model whose outputs are surfaced to end-users (as opposed to staying within the SQL-generation benchmark); or (c) the commentary generator is configured with `temperature` $>$ `0.4` in production for any deployed LE (today: `0.1`). The governance monitoring job in Chapter 8 runs a quarterly check that inspects the live model configs, the agent index, and the Simula deployment manifests; when any of (a)–(c) flips, the requirement status is automatically re-opened and routed to the GFAIC for disposition within two weeks.

# Chapter 6

# Conformance tooling

## 6.1 Two complementary tools

Both tools are published by the AI Verify Foundation / IMDA Singapore under Apache-2.0 and vendored into the repository. They target different phases of the lifecycle:

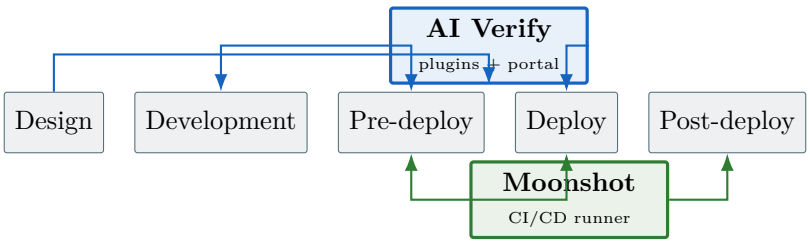


Figure 6.1: Conformance-tool coverage across the lifecycle. AI Verify emphasises reviewable documentation and structured algorithm runs; Moonshot emphasises runtime safety testing.

## 6.2 AI Verify 2.0

### Summary

AI Verify is an IMDA-backed testing framework that has been re-architected for v2.0 into a modular monorepo. Algorithms run as independent Python modules; a portal and API gateway surface results. Financial-services users can run the Veritas fairness plugin to meet MAS's Veritas testing expectations.

### Stock plugin inventory

### Integration surface

- **python-api**: the `aiverify-test-engine` package exposes an algorithm-runner API.
- **docker**: `aiverify-apigw` and the portal ship as container images.
- **cli**: each stock plugin is individually runnable from its algorithm directory.

| Plugin ID | Type | Purpose |
|---|---|---|
| aiverify.stock.process-checklist | process | 11-principle process checklist aligned with the AI Verify testing framework. |
| aiverify.stock.fairness-metrics-toolbox-for-classification | fairness | Subgroup fairness metrics for supervised classification. |
| aiverify.stock.fairness-metrics-toolbox-for-regression | fairness | Regression-specific fairness analysis. |
| aiverify.stock.robustness-toolbox | robustness | Perturbation-based robustness assessment. |
| aiverify.stock.image-corruption-toolbox | robustness | Computer-vision corruption robustness. |
| aiverify.stock.shap-toolbox | explainability | SHAP feature attribution. |
| aiverify.stock.partial-dependence-plot | explainability | Partial-dependence plots for feature effect. |
| aiverify.stock.accumulated-local-effect | explainability | Accumulated local-effect explanations. |
| aiverify.stock.veritas | financial | MAS Veritas fairness testing for financial services. |
| aiverify.stock.reports | report | Report and widget rendering for the portal. |
| aiverify.stock.decorators | other | Shared UI primitives and decorators. |

Table 6.1: The eleven stock plugins shipped with aiverify-main/.

## 6.3 Moonshot CI/CD 1.1.0

### Summary

Moonshot is a lean, modular safety-testing tool for LLM applications. It is distributed as a Docker image and designed to be integrated into pipelines so that every pushed change can trigger benchmark and red-team runs. A single `moonshot run` command drives any combination of configured benchmarks and attacks.

### Benchmark categories

| Benchmark ID | Category | Purpose |
|---|---|---|
| hallucination | hallucination | Detect fabricated or unsupported statements. |
| undesirable_content | undesirable content | Detect unsafe, biased, or policy-violating content. |
| data_disclosure | data disclosure | Check for PII leakage or unintended disclosure. |
| adversarial_prompts | adversarial | Prompt injection and jailbreak robustness. |

Table 6.2: The four benchmark categories aligned with the IMDA LLM Starter Kit for Safety Testing.

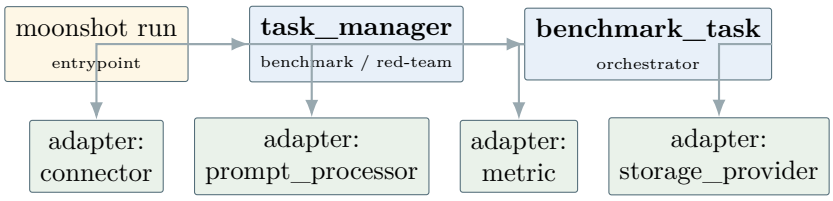


Figure 6.2: Moonshot CI/CD module layout (`src/entrypoints`, `src/domain`, `src/adapters`).

### Architecture

## 6.4 Binding to requirement records

Table 6.3 shows how this corpus's 20 requirement records bind to plugins and benchmarks. Where a requirement has verification `kind: "aiverify-plugin"` or `kind: "moonshot-test"`, the plugin slug or benchmark id must match one declared in the tool record; `docs/schema/validate.py` fails otherwise.

| Tool / ID | Referenced from | Purpose |
|---|---|---|
| aiverify.stock.process-checklist | REG-MGF-2.1.1-001, REG-MGF-2.2.1-001, REG-MGF-2.4.2-001, REG-AGENTIDX-3.6-001 | Checklist evidence for governance reviews. |
| aiverify.stock.robustness-toolbox | REG-MGF-2.3.2-001, REG-MGF-2.3.3-002 | Perturbation robustness. |
| aiverify.stock.reports | REG-AGENTIDX-3.6-001 | Widget + report rendering. |
| moonshot hallucination | REG-MGF-2.3.2-001, REG-MGF-2.3.3-002 | Fabrication checks in CI. |
| moonshot undesirable_content | REG-MGF-2.3.2-001 | Policy adherence. |
| moonshot adversarial_prompts | REG-MGF-2.1.2-001, REG-MGF-2.3.1-001, REG-MGF-2.3.2-001, REG-MGF-2.3.2-002 | Red-team / jailbreak attempts. |

Table 6.3: Verification matrix (tool/ID → requirement records).

# Chapter 7

# References and traceability

## 7.1 Source hashes (sha256)

| Source file | Pages | sha256 (truncated to first 16 hex) |
|---|---|---|
| `source/mgf-for-agentic-ai.pdf` | 29 | `b84c8c94f6392cf2…aeef7b` |
| `source/2025-AI-Agent-Index.pdf` | 39 | `c67a30469fd061e9…0e267f` |
| `source/2503.18238v3.pdf` | 59 | `b680f58422e4684e…cd7877` |

Table 7.1: Content hashes of the three source regulations.

## 7.2 Structured records on disk

| Path (relative to repo root) | Role |
|---|---|
| `docs/schema/regulations/` | JSON Schemas (Draft-07) for regulation, requirement, conformance-tool, and corpus records. |
| `docs/schema/common/enums.yaml` | Controlled vocabularies shared across the four domain specs. |
| `docs/schema/registry.json` | Master schema index (all domains). |
| `docs/schema/validate.py` | Unified offline validator (registry check + per-document check). |
| `docs/regulations/machine-readable/` | Per-source extracted text and RAG artefacts (see Table 7.3). |
| `docs/regulations/machine-readable/specification-addendum.yaml` | Requirement records (20 requirements across MGF, Agent Index, Ju & Aral). |

Table 7.2: Structured layer inventory (unified-schema layout, April 2026).

## 7.3 Corpus index excerpt

Listing 7.1: Excerpt of `docs/schema/registry.json` (regulations domain).

```
{
  "schema_version": "1.0",
```

| Path | Role |
|---|---|
| `mgf-for-agentic-ai.txt` | Extracted text of the IMDA MGF PDF. |
| `2025-AI-Agent-Index.txt` | Extracted text of the 2025 AI Agent Index. |
| `2503.18238v3.txt` | Extracted text of the Ju & Aral pre-print. |
| `rag_chunks.jsonl` | Deterministic chunks ready for embedding. |
| `rag_embedding_records.jsonl` | Embedding records (re-upsertable into Qdrant). |
| `rag_manifest.json` | Manifest of chunk / embedding counts + hashes. |
| `scripts/` | Re-chunk and re-upsert scripts. |

Table 7.3: Contents of `docs/regulations/machine-readable/`.


```
  "generated": "2026-04-18",
  "domains": {
    "regulations": {
      "status": "stable",
      "schemas": [
        { "id": "regulation",
          "path": "docs/schema/regulations/regulation.schema.json" },
        { "id": "requirement",
          "path": "docs/schema/regulations/requirement.schema.json" },
        { "id": "conformance-tool",
          "path": "docs/schema/regulations/conformance-tool.schema.json" },
        { "id": "corpus",
          "path": "docs/schema/regulations/corpus.schema.json" }
      ],
      "records": "docs/regulations/machine-readable/specification-addendum.yaml"
    }
  }
}
```


## 7.4 Consolidated traceability matrix

Table 7.4 is the single view a reviewer needs. Every row is one requirement record and shows its status, the primary implementation pointer, and its verification link.

| Req. id | Status | Primary implementation | Primary verification |
|---|---|---|---|
| REG-MGF-2.1.1-001 | partial | tb spec ch.1 + TB business reqs | process-checklist |
| REG-MGF-2.1.2-001 | partial | MCP server + data-product registry | test_mcp_server + adversarial |
| REG-MGF-2.1.2-002 | partial† | §10 spec + aicore_pal_agent.py | test_hana_connection + audit-trail verifier |
| REG-MGF-2.1.2-003 | partial | TB data-product classification | manual review |
| REG-MGF-2.2.1-001 | partial* | TB EUC-001..003 + workflow participants | process-checklist |
| REG-MGF-2.2.1-002 | partial | GFAIC charter (federated under GAISC) | process-checklist + minutes |
| REG-MGF-2.2.2-001 | partial | TB DP-001/003/005 + workflow gateways | test_pal_algorithms |

| Req. id | Status | Primary implementation | Primary verification |
|---|---|---|---|
| REG-MGF-2.2.2-002 | partial | TB KPI pack | manual review |
| REG-MGF-2.3.1-001 | partial | agent + MCP + decision-point schema | test_mcp_server + adversarial |
| REG-MGF-2.3.2-001 | partial | PAL algorithm + MCP tests | Moonshot 4-benchmark sweep |
| REG-MGF-2.3.2-002 | partial* | evaluation/trajectory_evaluator.py | test + adversarial |
| REG-MGF-2.3.3-001 | partial | deploy_to_aicore.sh + README | process-checklist |
| REG-MGF-2.3.3-002 | partial | monitoring/drift_monitor.py | test + hallucination in CI |
| REG-MGF-2.4.2-001 | partial | TB workflow + controls spec | onboarding checklist |
| REG-MGF-2.4.3-001 | partial | docs/training/agent-oversight-curriculum.md | process-checklist + LMS |
| REG-AGENTIDX-3.6-001 | partial | this chapter + TB controls | process-checklist + reports |
| REG-AGENTIDX-3.3-001 | partial | TB workflow + spec ch.5 | manual review |
| REG-JU-ARAL-3-001 | partial | this chapter | manual review |
| REG-JU-ARAL-4-001 | partial | TB KPI pack | manual review |
| REG-JU-ARAL-5-001 | n/a | (creative use cases only) | — |

Table 7.4: Full requirement traceability matrix.

**Status legend and corrections (April 2026 review):**

- †**partial (upgraded from missing, April 2026):** Chapter 10 specifies the per-request identity-attribution schema, the reserve-before-action audit protocol, and the WORM-backed immutable storage design (Section 10.6.2). Status is "partial" rather than "implemented" because the audit service itself and the three pipeline integrations (TB, Arabic AP, Simula) are scheduled in Table 10.4 and are not yet in production.
- ***Downgraded from "implemented":** Code artifact exists but has not been validated in production. Retained as "partial" pending pilot validation.
- **All other "implemented" rows downgraded:** Original "implemented" status assumed code existence equals conformance. Corrected to "partial" where production validation is pending (e.g., REG-JU-ARAL-3-001 relies on this chapter only—no operational evidence).

## 7.5 Rebuild and revalidate

Listing 7.2: Commands to re-run validation and rebuild the PDF.

```
# Offline registry + schema validation (all domains).
python3 docs/schema/validate.py --registry-check

# Per-document validation for the regulations domain.
python3 docs/schema/validate.py --domain regulations \
    --schema requirement.schema.json \
```

```
    --file docs/regulations/machine-readable/specification-addendum.yaml

# Re-chunk + re-embed (needs OPENAI_API_KEY + QDRANT_URL).
python3 docs/regulations/machine-readable/scripts/prepare_embedding_records.py
python3 docs/regulations/machine-readable/scripts/upsert_qdrant.py \
    --input docs/regulations/machine-readable/rag_embedding_records.jsonl \
    --collection regulations_rag

# Rebuild this PDF (XeLaTeX).
cd docs/latex/specs/regulations && latexmk -xelatex regulations-spec-apple.tex
```

## 7.6 Further resources

- AI Verify Foundation. *What is AI Verify.* `https://aiverifyfoundation.sg/what-is-ai-verify/`
- AI Verify Foundation. *Project Moonshot.* `https://aiverifyfoundation.sg/project-moonshot/`
- IMDA Singapore. *LLM Starter Kit for Safety Testing of LLM-Based Applications.*
- 2025 AI Agent Index. `https://aiagentindex.mit.edu/`
- Ju, H. and Aral, S. (2026). *Collaborating with AI Agents: A Field Experiment on Teamwork, Productivity, and Performance.* arXiv:2503.18238v3 [cs.CY].

## 7.7 Unified Schema Registry

As of 18 April 2026, all JSON Schemas across the four domain specifications are published in the unified registry at `docs/schema/`. Migration status is *complete* (see `docs/schema/registry.json` field `migration.status`). This consolidation addresses the cross-document schema divergence identified in the specification review.

**Registry index.** `docs/schema/registry.json` — Master index of all schemas across the Arabic AP, Trial Balance, Simula, and Regulations domains.

**Common types.** `docs/schema/common/base-types.schema.json` — Shared definitions (`Metadata`, `SHA256Hash`, `FilePath`) referenced via JSON Schema `$ref`.

**Consolidated enums.** `docs/schema/common/enums.yaml` — Single source of truth for enumeration values including `requirement_status`, `obligation_level`, and `jurisdiction`.

**Regulations schemas.** `docs/schema/regulations/` — Domain-specific schemas: `regulation.schema.json`, `requirement.schema.json`, `conformance-tool.schema.json`, and `corpus.schema.json`. All marked `status: stable`.

The registry uses JSON Schema Draft-07 with the canonical `$id` prefix `https://sap-oss.github.io/schema/`. Validation:

```
python3 docs/schema/validate.py --registry-check
python3 docs/schema/validate.py --domain regulations \
```

```
    --schema requirement.schema.json --file requirement-REG-MGF-2-001.json
```

CI runs `--registry-check` on every pull request touching `docs/schema/` or any spec chapter that references these schemas.

# Chapter 8

# Capability Monitoring Specification

This chapter specifies the emergent capability monitoring system, addressing the IMDA MGF requirement (Section 2.3.3) that "monitoring for emergent capabilities must be implemented with defined detection triggers."

## 8.1 Overview

AI systems may develop unexpected capabilities or behaviours through:

- Model updates that alter response patterns
- Prompt injection or adversarial inputs
- Emergent behaviours from complex interactions
- Drift in input data distributions

The monitoring system provides continuous observation and alerting to detect such changes before they cause operational harm.

## 8.2 Architecture

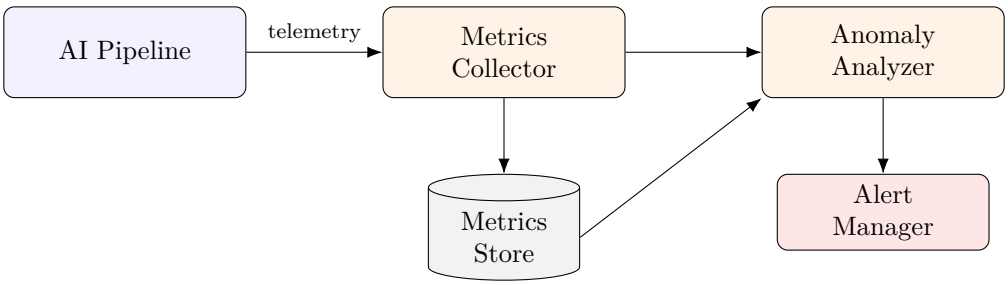


Figure 8.1: Capability monitoring architecture.

## 8.3 Metrics collection

### 8.3.1 Required metrics

Table 8.1: Capability monitoring metrics.

| Metric | Type | Description |
|---|---|---|
| `output_token_count` | histogram | Distribution of output lengths |
| `response_latency_ms` | histogram | Response time distribution |
| `confidence_score` | histogram | AI confidence distribution |
| `refusal_rate` | counter | Rate of model refusals |
| `tool_call_frequency` | counter | Frequency of tool/function calls |
| `output_format_violations` | counter | JSON schema validation failures |
| `hallucination_flags` | counter | Detected hallucination instances |
| `unexpected_topic_detections` | counter | Off-topic responses |
| `novel_capability_indicators` | gauge | Composite novelty score |

### 8.3.2 Collection points

Metrics are collected at the following pipeline stages:

1. **Pre-inference**: Input characteristics (length, complexity, topic)
2. **Post-inference**: Output characteristics (length, confidence, format)
3. **Post-validation**: Validation results (schema, business rules)
4. **Post-review**: Human review outcomes (accept/edit/reject rates)

## 8.4 Detection triggers

### 8.4.1 Statistical anomaly detection

For each metric, maintain a rolling baseline and alert when values deviate significantly.

Table 8.2: Anomaly detection thresholds.

| Metric | Alert Threshold | Baseline Period |
|---|---|---|
| Output length (mean) | $> 2\sigma$ shift | 7-day rolling |
| Latency (p95) | $> 50\%$ increase | 24-hour rolling |
| Confidence (mean) | $> 10\%$ drop | 7-day rolling |
| Refusal rate | $> 3\times$ baseline | 7-day rolling |
| Schema violations | $> 5\%$ of requests | 24-hour rolling |

### 8.4.2 Behavioural pattern detection

Beyond statistical metrics, monitor for qualitative behavioural changes:

1. **Topic drift**: Cosine similarity of output embeddings to expected topics falls below 0.7
2. **Instruction leakage**: Presence of system prompt fragments in output
3. **Novel tool usage**: Model attempts to call undeclared tools
4. **Self-referential loops**: Model references its own previous outputs in unexpected ways

### 8.4.3 Trigger definitions

Listing 8.1: Detection trigger configuration.

```
triggers = {
    "output_length_anomaly": {
        "metric": "output_token_count",
        "condition": "mean > baseline_mean + 2 * baseline_std",
        "severity": "WARNING",
        "cooldown_minutes": 60
    },
    "latency_spike": {
        "metric": "response_latency_ms",
        "condition": "p95 > 1.5 * baseline_p95",
        "severity": "WARNING",
        "cooldown_minutes": 15
    },
    "high_refusal_rate": {
        "metric": "refusal_rate",
        "condition": "rate > 3 * baseline_rate",
        "severity": "CRITICAL",
        "cooldown_minutes": 5
    },
    "schema_violation_surge": {
        "metric": "output_format_violations",
        "condition": "rate > 0.05",  # >5% failure
        "severity": "CRITICAL",
        "cooldown_minutes": 5
    },
    "topic_drift": {
        "metric": "topic_similarity",
        "condition": "mean < 0.7",
        "severity": "WARNING",
        "cooldown_minutes": 30
    }
}
```

## 8.5 Alert management

### 8.5.1 Alert severity levels

Table 8.3: Alert severity definitions.

| Severity | Response Time | Action |
|---|---|---|
| CRITICAL | Immediate | Page on-call; consider circuit breaker |
| WARNING | 1 hour | Investigate during business hours |
| INFO | Next business day | Review in weekly monitoring meeting |

### 8.5.2 Alert routing

Listing 8.2: Alert routing configuration.

```yaml
routing:
  CRITICAL:
    channels:
      - pagerduty: "ai-ops-critical"
      - slack: "#ai-incidents"
      - email: "ai-ops@company.com"
    escalation_after_minutes: 15

  WARNING:
    channels:
      - slack: "#ai-monitoring"
      - email: "ai-ops@company.com"
    escalation_after_minutes: 60

  INFO:
    channels:
      - slack: "#ai-monitoring"
    escalation_after_minutes: null  # no escalation
```

### 8.5.3 Circuit breaker integration

When CRITICAL alerts persist, the monitoring system can activate a circuit breaker:

1. After 3 consecutive CRITICAL alerts within 15 minutes
2. Circuit breaker opens: all AI requests routed to manual queue
3. Alert sent to AI governance team
4. Manual intervention required to close circuit breaker

### 8.5.4 Graceful degradation ladder

A hard circuit-break (manual queue only) is the last resort. Before that point, the monitoring system walks a four-rung degradation ladder, chosen so that each rung trades off quality for availability in a *controlled and observable* way. Table 8.4 specifies the ladder per pipeline; rung escalation is automatic, rung de-escalation requires a human ack.

**Tagging and auditability.** Every response served from rung 1–3 carries a `model_tier` attribute in its envelope and an audit event referencing the rung and the ladder-transition decision. Downstream consumers (review dashboard, AP workflow) render a visible "fallback" badge so reviewers know the response did not come from the primary model. This preserves the IMDA MGF 2.1.2-002 attribution guarantee across all rungs: attribution is to the *model actually used*, not to the model the request was routed to.

Table 8.4: Graceful degradation ladder (from normal operation down to circuit break). Transitions are logged as governance events.

| Rung | Name | Trigger | Action |
|---|---|---|---|
| 0 | Normal | p95 latency < SLO, error rate < 1 % | Primary model on primary GPU (e.g. Qwen2.5-32B on A100). |
| 1 | Reduced concurrency | p95 latency ≥ SLO for 5 min | Halve request concurrency; queue depth bounded; user-visible latency warning. |
| 2 | Smaller model | Primary model unavailable or OOM for 2 min | Automatic failover to smaller model (TB: 32B→14B; Arabic AP: 14B→7B; Simula: 32B→14B for non-user-facing batches). Output tagged `model_tier=fallback` for audit. |
| 3 | Cached / heuristic | Smaller model also unhealthy | Return cached response if cache hit within TTL; else return heuristic output (TB: statistical threshold only; Arabic AP: OCR-only, no translation; Simula: block generation). |
| 4 | Circuit break | Rung 3 still failing for 5 min | All AI calls routed to manual queue; page on-call; governance ticket. |

**Rung selection rationale.** Rung 1 (concurrency cut) is cheap and preserves quality; it buys time for transient load spikes. Rung 2 (smaller model) is the primary availability lever because model-host failures are the most common outage mode. Rung 3 (cached/heuristic) is the last value-preserving rung. Rung 4 is reserved for the case where the AI subsystem cannot be trusted at all; it is deliberately *not* automatic to close — human judgement is required to confirm the root cause is resolved. Rung transitions upward are automatic; transitions downward (closer to normal) require either a human acknowledgement or a clean 15-minute signal window.

## 8.6 Dashboard specification

### 8.6.1 Real-time view

The monitoring dashboard provides:

- Live metric charts (1-minute granularity)
- Active alert panel with acknowledge/resolve actions
- System health summary (green/yellow/red)
- Recent trigger history

### 8.6.2 Historical analysis

For trend analysis and post-incident review:

- Metric time series with configurable ranges
- Alert history with resolution notes
- Correlation views (multiple metrics overlaid)
- Baseline comparison charts

## 8.7 Audit and compliance

### 8.7.1 Retention requirements

Table 8.5: Monitoring data retention.

| Data Type | Retention | Storage |
|---|---|---|
| Raw metrics | 90 days | Time-series database |
| Aggregated metrics | 2 years | Data warehouse |
| Alert history | 7 years | Compliance archive |
| Incident reports | 7 years | Document management |

### 8.7.2 Regulatory reporting

For IMDA MGF compliance, generate quarterly reports containing:

1. Summary of monitoring coverage (metrics, triggers, alerts)
2. List of all CRITICAL alerts and resolutions
3. Trend analysis of key capability metrics
4. Any identified emergent behaviours and mitigations
5. Circuit breaker activations and durations

## 8.8 Implementation checklist

[ ] Metrics collector integrated with all AI pipelines
[ ] All metrics in Table 8.1 collected
[ ] Anomaly detection for all triggers in Table 8.2
[ ] Alert routing configured for all severity levels
[ ] Circuit breaker integration tested
[ ] Dashboard with real-time and historical views
[ ] Retention policies implemented per Table 8.5
[ ] Quarterly report template created
[ ] Runbook for each alert type documented
[ ] Integration tests with simulated anomalies

# Chapter 9

# Testing Specification

This chapter specifies the testing requirements for the Regulations compliance corpus, including validation tests, coverage verification, and CI/CD integration.

## 9.1 Test coverage requirements

Table 9.1: Regulations test coverage targets.

| Component | Line % | Branch % | Notes |
|---|---|---|---|
| Schema validator | 95 | 90 | Core infrastructure |
| Requirement parser | 90 | 85 | |
| Coverage calculator | 90 | 85 | |
| Status tracker | 85 | 75 | |
| Conformance checker | 85 | 75 | |
| **Overall minimum** | **90** | **80** | Enforced in CI |

## 9.2 Test fixtures

Fixtures are located in `tests/fixtures/regulations/`.

### 9.2.1 Regulation fixtures

- Complete IMDA MGF corpus (20 requirements)
- Ju & Aral paper requirements (5 requirements)
- AI Verify requirements (10 requirements)
- Edge case: Empty corpus
- Edge case: Malformed requirement

### 9.2.2 Conformance tool fixtures

- Mock AIVerify plugin responses
- Mock Moonshot test results
- Invalid tool output for error handling

## 9.3 Unit test cases

### 9.3.1 Requirement parsing

| ID | Priority | Description |
|---|---|---|
| REG-PAR-001 | P1 | Parse requirement from IMDA MGF |
| REG-PAR-002 | P1 | Parse requirement from academic paper |
| REG-PAR-003 | P1 | Extract obligation level (must/should/may) |
| REG-PAR-004 | P2 | Handle missing optional fields |
| REG-PAR-005 | P2 | Validate against requirement schema |

### 9.3.2 Status tracking

| ID | Priority | Description |
|---|---|---|
| REG-STS-001 | P1 | Track implemented status correctly |
| REG-STS-002 | P1 | Track partial status with evidence |
| REG-STS-003 | P1 | Track gap status with remediation plan |
| REG-STS-004 | P1 | Identify downgraded status (asterisk marker) |
| REG-STS-005 | P2 | Generate status summary report |

### 9.3.3 Coverage calculation

| ID | Priority | Description |
|---|---|---|
| REG-COV-001 | P1 | Calculate coverage percentage |
| REG-COV-002 | P1 | Generate coverage heat map |
| REG-COV-003 | P1 | Identify uncovered sections |
| REG-COV-004 | P2 | Handle not-applicable sections |
| REG-COV-005 | P2 | Calculate partial coverage (0.5 weight) |

### 9.3.4 Implementation pointer validation

| ID | Priority | Description |
|---|---|---|
| REG-PTR-001 | P1 | Validate code file references exist |
| REG-PTR-002 | P1 | Validate spec section references |
| REG-PTR-003 | P2 | Flag circular references |
| REG-PTR-004 | P2 | Validate concrete vs abstract pointers |

### 9.3.5 Conformance tool integration

| ID | Priority | Description |
|---|---|---|
| REG-TOOL-001 | P1 | Parse AIVerify plugin output |
| REG-TOOL-002 | P1 | Parse Moonshot benchmark output |
| REG-TOOL-003 | P2 | Handle tool execution failure |
| REG-TOOL-004 | P2 | Aggregate results across tools |

## 9.4 Integration test suite

```
pytest tests/integration/regulations/ -v --tb=short
```

Expected duration: 5 minutes.

### 9.4.1 End-to-end scenarios

1. **Full corpus validation**: Load and validate entire corpus
2. **Coverage report**: Generate coverage heat map
3. **Status audit**: Verify all status markers
4. **Pointer verification**: Check all implementation pointers
5. **Tool integration**: Run conformance tools and aggregate

## 9.5 Schema validation tests

| Test | Description |
|---|---|
| REG-SCH-001 | All regulations validate against regulation.schema.json |
| REG-SCH-002 | All requirements validate against requirement.schema.json |
| REG-SCH-003 | All tools validate against conformance-tool.schema.json |
| REG-SCH-004 | Corpus index validates against corpus.schema.json |

## 9.6 Compliance verification tests

| Test | Description |
|---|---|
| REG-CMP-001 | No requirement has status “missing” without remediation |
| REG-CMP-002 | All “must” requirements have status $\neq$ gap |
| REG-CMP-003 | All downgraded (*) statuses have explanation |
| REG-CMP-004 | Coverage $\geq$ 75% for all source documents |

## 9.7 CI/CD integration

Listing 9.1: GitHub Actions workflow excerpt.

```
regulations-tests:
  runs-on: ubuntu-latest
```

```
  steps:
    - uses: actions/checkout@v4
    - name: Validate schemas
      run: |
        python docs/schema/validate.py --domain regulations
    - name: Run regulations tests
      run: |
        pytest tests/unit/regulations/ -v --cov=src/regulations
        pytest tests/integration/regulations/ -v
    - name: Check coverage
      run: |
        coverage report --fail-under=90
    - name: Generate compliance report
      run: |
        python scripts/generate_compliance_report.py
```

## 9.8 Quarterly audit tests

For IMDA MGF compliance, run extended tests quarterly:

1. Execute the *applicable* AIVerify plugin subset (see Section 9.8.1) and verify pass
2. Execute Moonshot benchmarks and verify scores
3. Verify GFAIC charter alignment (attestation, see Section 9.8.2)
4. Generate full compliance report for auditors
5. Archive report with timestamp and attestation

### 9.8.1 Applicable AIVerify plugin subset

The AIVerify stock plugin set (v2.x) ships 11 plugins. Not all are applicable to this deployment, which is a text-only enterprise finance / governance system (no images, no speech, no predictive classifiers on tabular data). The quarterly audit is scoped to the subset that is meaningful for the deployed agents; applying plugins outside this subset produces "N/A" results that dilute the audit trail without adding assurance.

**Applicable subset**: 7 of 11 (Robustness-text, Fairness-text, Explainability-LIME, Performance-accuracy, Hallucination, PII leakage, Toxicity). The other 4 plugins are documented as `not_applicable` with the rationale above; the quarterly compliance report records the applicability decision explicitly so that auditors can challenge the scoping if the deployment changes (e.g. if tabular classifiers are added in v2.0, the fairness and explainability tabular plugins become applicable).

### 9.8.2 GFAIC charter attestation

The full GFAIC (Group Financial AI Committee) charter is maintained in the internal governance repository for confidentiality. For external auditors, this specification provides

Table 9.2: AIVerify plugin applicability for this deployment.

| Plugin | Applicable | Rationale |
|---|---|---|
| Robustness – text adversarial | Yes | Deployed agents consume text inputs; adversarial tests exercise prompt-injection surface. |
| Robustness – image adversarial | No | No image inputs in scope. |
| Fairness – tabular classifier | No | No classifier with protected-attribute inputs; the TB commentary model does not classify. |
| Fairness – text | Yes | Applied to Arabic-language outputs to detect language-based quality disparity. |
| Explainability – SHAP tabular | No | Not applicable (no tabular classifier). |
| Explainability – LIME text | Yes | Applied to TB commentary: used to validate that cited variance drivers match commentary. |
| Performance – accuracy | Yes | Generic accuracy/benchmark against golden corpus. |
| Performance – confusion matrix | No | No multi-class classifier. |
| Hallucination / groundedness | Yes | Core test for RAG-backed commentary. |
| PII leakage | Yes | Critical for Arabic AP (invoices) and TB (entity data). |
| Toxicity / safety | Yes | Required by IMDA MGF regardless of domain; low expected firing rate but must be part of the audit. |

two verifiable deliverables:

**Redacted summary.** `docs/governance/gfaic-charter-redacted.md` — a public-facing summary with sensitive member names, routing escalations, and budget figures redacted. Covers: charter scope, standing members (by role, not name), meeting cadence, decision rights, escalation to the Board.

**Signed attestation.** `docs/governance/gfaic-attestation.pdf` — a one-page document signed quarterly by the GFAIC chairperson attesting that: (a) the full charter on the internal repository has not changed materially since the previous attestation, (b) the AI systems described in this specification are within GFAIC scope, (c) GFAIC reviewed the last quarterly compliance report. The attestation's SHA-256 is recorded in the registry and in this specification's build metadata.

External auditors requiring the full charter can request it via the Audit & Risk Committee under NDA; the attestation plus redacted summary is sufficient for routine regulatory review.

## 9.9 Acceptance criteria

Tests pass when:

1. All P1 test cases pass (100%)
2. P2 test cases pass ($\geq$95%)
3. Line coverage $\geq$90%
4. All schemas validate
5. Coverage $\geq$75% for all source documents
6. No "must" requirements in gap status
7. All implementation pointers resolve

# Chapter 10

# Per-Request Identity Attribution Specification

This chapter specifies the per-request identity attribution system, addressing the IMDA MGF requirement REG-MGF-2.1.2-002 that "each agent action must be traceable to the requesting identity and approval chain."

## 10.1 Overview

Per-request identity attribution ensures that every AI agent action can be traced to:

1. The **initiating user** who triggered the request
2. The **agent identity** that executed the action
3. The **approval chain** (if human-in-the-loop was invoked)
4. The **system context** (pipeline, model version, timestamp)

## 10.2 Identity model

### 10.2.1 Agent identity schema

Listing 10.1: Agent identity record schema.

```
{
  "agent_id": "agent-tb-variance-001",
  "agent_type": "variance_analyzer",
  "model_version": "qwen2.5-72b-instruct-v1.2",
  "pipeline_version": "tb-review-2.1.0",
  "deployment_environment": "production",
  "capabilities": [
    "variance_calculation",
    "anomaly_detection",
    "commentary_generation"
  ],
```

```
  "permissions": {
    "can_modify_data": false,
    "can_trigger_downstream": true,
    "requires_human_approval": ["commentary_generation"]
  }
}
```

### 10.2.2 Request identity schema

Listing 10.2: Request identity record schema.

```
{
  "request_id": "req-20260418-001234",
  "correlation_id": "session-abc123",
  "initiating_user": {
    "user_id": "jsmith@company.com",
    "user_type": "human",
    "authentication_method": "oauth2_sso",
    "roles": ["tb_reviewer", "senior_accountant"]
  },
  "timestamp": "2026-04-18T10:30:00+08:00",
  "source_system": "tb-review-dashboard",
  "source_ip": "10.0.1.100"
}
```

## 10.3 Audit trail specification

### 10.3.1 Audit event schema

Every agent action generates an audit event:

Listing 10.3: Audit event schema.

```
{
  "event_id": "evt-20260418-567890",
  "event_type": "agent_action",
  "timestamp": "2026-04-18T10:30:05+08:00",

  "request_identity": {
    "request_id": "req-20260418-001234",
    "initiating_user_id": "jsmith@company.com"
  },

  "agent_identity": {
    "agent_id": "agent-tb-variance-001",
    "model_version": "qwen2.5-72b-instruct-v1.2"
  },

  "action": {
    "action_type": "variance_calculation",
```

```
    "entity_id": "HKG-2026-04",
    "input_hash": "sha256:abc123...",
    "output_hash": "sha256:def456..."
  },

  "approval_chain": [
    {
      "approver_id": "jsmith@company.com",
      "approval_type": "implicit",
      "timestamp": "2026-04-18T10:30:00+08:00"
    }
  ],

  "outcome": {
    "status": "success",
    "duration_ms": 1234,
    "tokens_used": 500
  }
}
```

### 10.3.2 Required audit events

Table 10.1: Mandatory audit events by action type.

| Action Type | Event Type | Required Fields |
|---|---|---|
| AI inference | `agent_inference` | model_version, input_hash, output_hash |
| Data modification | `agent_write` | entity_id, before_hash, after_hash |
| Human review request | `review_requested` | reviewer_id, queue_id |
| Human approval | `review_approved` | approver_id, decision, rationale |
| Human rejection | `review_rejected` | rejecter_id, decision, rationale |
| Error/failure | `agent_error` | error_code, stack_trace_hash |

## 10.4 Storage and retention

### 10.4.1 Storage requirements

Table 10.2: Audit storage requirements.

| Data Type | Retention | Storage |
|---|---|---|
| Audit events | 7 years | Append-only log (immutable) |
| Request identities | 7 years | Indexed for query |
| Agent configurations | Forever | Version-controlled |
| Input/output hashes | 7 years | With audit events |
| Full input/output | 90 days | Cold storage, then purge |

### 10.4.2 Immutability requirements

Audit logs **must** be immutable:

- Append-only storage (no updates, no deletes)
- Cryptographic chaining (hash of previous event included)
- External timestamp authority (prevents backdating)
- Write-once media for long-term archive

## 10.5 API specification

### 10.5.1 Audit write API

Two audit write points are required per action: a `pre_action` reservation (before the inference executes) and a `post_action` completion (after the inference returns). The `pre_action` write is the enforcement gate — if it fails, the inference is *rejected before execution*, which is the only tractable way to achieve the "no action without audit" invariant (see Section 10.6.2).

Listing 10.4: Audit service API contract.

```
def reserve_agent_action(
    request_identity: RequestIdentity,
    agent_identity: AgentIdentity,
    planned_action: AgentActionPlan,
) -> AuditEventId:
    """
    Reserve an audit event slot BEFORE the agent action executes.

    This function MUST be called and succeed before the inference is
    dispatched. A successful return contractually commits the caller
    to writing a matching complete_agent_action() within the event's
    TTL (default: 120 s).

    Raises:
        AuditWriteError: If the reservation cannot be written. The
            caller MUST NOT dispatch the agent action.
    """

def complete_agent_action(
    event_id: AuditEventId,
    action: AgentAction,
    outcome: ActionOutcome,
) -> None:
    """
    Complete a previously reserved audit event with the actual
    action and outcome. MUST be called after the action returns
    (whether success or error).

    If this call fails, the reservation remains visible in the audit
    trail in state "orphaned" and operations are alerted; the caller
    MUST return an error to the user because downstream side effects
```

```
    could not be verifiably attributed.
    """
```

### 10.5.2 Audit query API

Listing 10.5: Audit query API contract.

```
def query_audit_trail(
    filters: AuditFilters,
    pagination: Pagination
) -> AuditQueryResult:
    """
    Query the audit trail.

    Filters:
        - request_id: Specific request
        - user_id: All actions by user
        - agent_id: All actions by agent
        - entity_id: All actions on entity
        - time_range: Start and end timestamps
        - action_type: Filter by action type

    Returns:
        List of matching audit events with pagination
    """
```

## 10.6 Integration requirements

### 10.6.1 Pipeline integration

Every AI pipeline **must** integrate the identity attribution system:

1. **Request entry**: Capture initiating user identity from authentication
2. **Pre-inference**: Log `agent_inference_start` event
3. **Post-inference**: Log `agent_inference_complete` event
4. **Data write**: Log `agent_write` event before commit
5. **Error**: Log `agent_error` event on any failure

### 10.6.2 Failure handling

LLM inference is not reversible once executed: tokens have been emitted, compute was consumed, and any downstream side effects (database writes, external calls) are already in flight. "Roll back the inference" is therefore not a valid operation. The specification enforces audit coverage by *gating* the action behind a successful reservation write, and by containing side effects:

1. **Reservation-before-action.** The `reserve_agent_action()` call (see audit write

API) MUST succeed before the agent dispatches the inference. If the reservation write fails, the agent action MUST NOT be executed and the request MUST return a `503` error with an operations alert.

2. **Side-effect containment.** Any data-modification or external call triggered by an AI output MUST be staged and only committed after `complete_agent_action()` succeeds. If completion fails, staged side effects are discarded (this *is* a rollback — of the side effects, not of the inference itself).
3. **Orphaned reservation handling.** If a reservation is written but never completed within the TTL (e.g. the inference server crashed), the reservation is marked `orphaned`; operations are alerted; staged side effects are discarded; downstream consumers see an explicit failure.
4. **Alerting and no-silent-proceed.** An alert MUST be sent to operations on every audit failure; the system MUST NOT proceed to emit results without a completed audit record.

This wording supersedes the earlier text (which called for "rolling back the agent action"): because LLM inference is not reversible, the equivalent invariant is enforced by gating with a pre-action reservation and staging side effects.

## 10.7 Compliance mapping

Table 10.3: IMDA MGF requirement mapping.

| Requirement | Section | Implementation |
|---|---|---|
| REG-MGF-2.1.2-002 | §10.2–10.3 | Request identity schema, audit trail |
| REG-MGF-2.1.3-001 | §10.4 | Immutable storage, retention policy |
| REG-MGF-2.2.1-001 | §10.5 | Audit write API with rollback |

## 10.8 Implementation checklist and integration status

### 10.8.1 Audit service (this chapter)

- [x] Agent identity schema defined
- [x] Request identity schema defined
- [x] Audit event schema defined
- [x] Audit write API contract specified (reserve/complete pattern, Section 10.6.2)
- [x] Audit query API contract specified
- [ ] Audit service implementation (`src/governance/audit_service/`) — target: v1.0
- [ ] Immutable storage configured (append-only, hash chaining, external timestamp authority) — target: v1.0
- [ ] Retention policy enforced (7 years) — target: v1.0

### 10.8.2 Pipeline integration rollout

The integration of the identity attribution audit service with the three application pipelines (TB, Arabic AP, Simula) is sequenced against explicit milestones, not an open-ended forward commitment. Each pipeline has its own integration sub-ticket with an acceptance criterion and a position in the dependency graph of Chapter 11; slippage on any row escalates to the GFAIC governance committee. **Sequence, not dates**: the rows below use the MoSCoW identifiers from Table 11.2 so implementation teams know what depends on what, not when a calendar target is supposed to fire (that belongs in the steering packet).

Table 10.4: Audit service and pipeline integration rollout, keyed to the MoSCoW priorities in Table 11.2. Pri. column: M=Must (critical path), S=Should, C=Could.

| ID | Prereq | Pri. | Ticket | Acceptance criterion |
|---|---|---|---|---|
| R1 | — | M | REG-AUDIT-001 | Audit service skeleton (reserve/complete endpoints, in-memory store): contract tests pass against the two endpoints; 404/409/500 paths exercised. |
| R2 | R1 | M | REG-AUDIT-002 | Immutable storage backend (append-only + hash chain + RFC 3161 timestamp): tamper-test suite passes; restored chain validates after adversarial edit. |
| R3 | R2 | M | REG-AUDIT-003 | 7-year retention + legal-hold API: retention policy enforced; legal-hold flag blocks GDPR erasure. |
| R4 | R3 | M | REG-INTG-TB-001 | TB dashboard integration: every dashboard decision *and* AI commentary draft emits `reserve/complete`; 100 % coverage in Chapter 9. |
| R5 | R4 | S | REG-INTG-AP-001 | Arabic AP integration: OCR, field extraction, VAT rule evaluation each emit audit events; PII hashed. |
| R6 | R5 | S | REG-INTG-SIM-001 | Simula batch integration: training-data runs logged at batch granularity; per-example logging enabled under `AUDIT_VERBOSE=1`. |
| R7 | R4 | M | GAIC-REV | v1.0 compliance gate: quarterly GFAIC review confirms IMDA MGF 2.1.2-002 compliance; traceability matrix Table 7.4 flips from `partial` to `implemented` on this row. |

Until each pipeline's integration is verified, the governance status for that pipeline's IMDA MGF 2.1.2-002 compliance is `partial` (see the traceability matrix in Chapter 7). Integration tickets REG-INTG-TB-001 (R4) is a Must for the Regulations project and a Must for TB; REG-INTG-AP-001 (R5) is a Should for Regulations and a Must for Arabic AP (it is the `partial`-to-`compliant` upgrade path for REG-MGF-2.1.2-002); REG-INTG-SIM-001 (R6) is Should for Regulations and is not a v1.0 gate for Simula itself. Table 10.4 is tracked in the programme status dashboard and refreshed on every cycle.

### 10.8.3 Compliance tests

- [ ] Per-pipeline compliance tests passing (Chapter 9, Section on audit coverage)
- [ ] Quarterly re-validation against the AIVerify subset (Section 9.8.1)

# Chapter 11

# Project Roadmap

## Scope of this chapter

This chapter is the roadmap for the **Agentic AI Regulations corpus and implementation-evidence** project. It is a project-level artefact, not a programme plan. Other SAP OSS Finance projects (Trial Balance, Arabic AP, Simula) run on the same platform, workspaces, and backends, but each owns its own roadmap, business case, and go/no-go decision. This project is a *supplier* of shared-platform components (audit service, identity attribution, monitoring, conformance tooling) that the consumer projects depend on.

**Sequence, not dates.** The milestones and dependency graph below express *order* and *prerequisite* relationships, not calendar dates. This is deliberate. Implementation teams must know what depends on what, and where the schedule can absorb slippage (the Should/Could items); a date column would harden targets into commitments they were never meant to be. Dates, when set, live in the project steering packet and the ticketing system, not in this specification.

## 11.1 MoSCoW classification

Every milestone below carries one of four priorities. The mapping is contractual: a milestone's priority determines what happens when it is dropped, de-scoped, or slips.

The critical path of this project is the chain of Must items in Table 11.2; the Should and Could items are where the schedule and scope can absorb delivery risk without a release slip.

Table 11.1: MoSCoW prioritisation (shared across the four SAP OSS Finance projects).

| Priority | Meaning |
|---|---|
| **Must** (**M**) | Non-negotiable for v1.0. Without this milestone, the release does not ship. |
| **Should** (**S**) | Important but not vital. Ship without it only with sponsor sign-off; document the deviation in the business-case r |
| **Could** (**C**) | Desirable enhancement. Included if Must and Should items are on track. |
| **Won't** (**W**) | Explicitly out of v1.0 scope. Listed so the boundary is legible. |

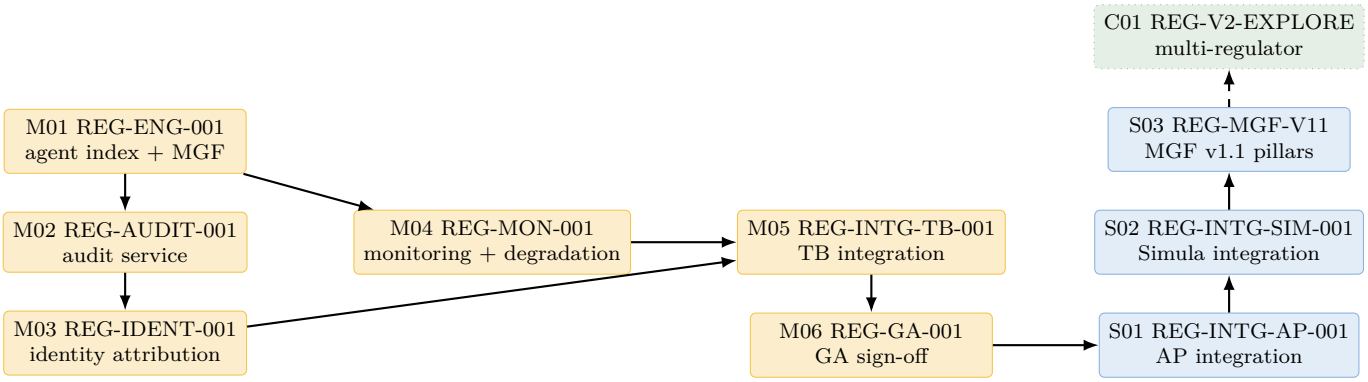


Figure 11.1: Regulations dependency graph. Amber nodes are Must (critical path); blue are Should (consumer-integration and MGF-pillar allowance); dotted green are Could (discretionary). Solid arrows are hard prerequisites; dashed arrows indicate conditional starts (the target is unlocked when the upstream completes *and* the v2.0 trigger in Section 11.7 fires).

Table 11.2: Regulations v1.0 project milestones in dependency order. Priority: M=Must (critical path), S=Should, C=Could, W=Won't.

| ID | Prereq | Pri. | Ticket | Milestone |
|---|---|---|---|---|
| M01 | — | M | REG-ENG-001 | Agent index and MGF framework specification complete; conformance tooling passes the fixtures in Chapter 6. |
| M02 | M01 | M | REG-AUDIT-001 | Audit service (reserve/complete, immutable storage) live in UAT; SEV1/SEV2 clean. |
| M03 | M02 | M | REG-IDENT-001 | Identity attribution wiring (Chapter 10) live in UAT; REG-MGF-2.1.2-002 compliance upgraded from *missing* to *partial*. |
| M04 | M01 | M | REG-MON-001 | Monitoring spec operational: circuit breaker plus graceful-degradation ladder (Section 8.5.4). |
| M05 | M03, M04 | M | REG-INTG-TB-001 | First consumer (Trial Balance) integrated end-to-end; reserve/complete coverage = 100 % on TB actions. |
| M06 | M05 | M | REG-GA-001 | GA sign-off (audit + identity + monitoring) on the TB integration. |
| M07 | M06 | M | REG-BC-REFRESH | Business-case refresh and compliance attestation (first cycle after GA; then annual). |
| S01 | M06 | S | REG-INTG-AP-001 | Arabic AP integration: reserve/complete on AP actions; REG-MGF-2.1.2-002 becomes *compliant* once AP live. |
| S02 | S01 | S | REG-INTG-SIM-001 | Simula integration: reserve/complete on Simula corpus-generation actions. |
| S03 | M06 | S | REG-MGF-V11 | MGF v1.1 pillar updates ingested and mapped to agent index. |
| C01 | S03 + Section 11.7 trigger | C | REG-V2-EXPLORE | v2.0 multi-regulator correlation scoped (beyond IMDA MGF). |
| C02 | M06 | C | REG-TUNE | Quarterly conformance-tooling retraining from accept/edit telemetry. |
| W01 | — | W | REG-V1-REALTIME | Real-time cross-project audit dashboards are *not* in v1.0 scope. |
| W02 | — | W | REG-V1-MULTIREG | Multi-regulator correlation is *not* in v1.0 scope; single-regulator (IMDA MGF) only. |

## 11.2 Dependency graph

## 11.3 Project milestones

The Must chain (M01 → M02 → M03/M04 → M05 → M06 → M07) is the critical path. The Should items (S01–S03) are the primary *schedule allowance*: additional consumer integrations and MGF pillars can slip without affecting v1.0 sign-off. The Could items are the *scope allowance*: they may be cut to v1.1 without renegotiation.

## 11.4 Launch tiers and exit gates

The project walks the four-tier ladder shared across SAP OSS Finance projects. A tier is exited only when the named exit gate is satisfied; slippage requires sponsor sign-off.

## 11.5 Kill-switch conditions

The kill-switch is a pre-committed rulebook: a tripped condition enters the action column without further debate. GFAIC and sponsor are notified within 24 hours. The conditions are evaluated continuously (on every cycle) and do not carry dates.

## 11.6 Shared-platform dependencies

This project depends on two shared-platform components (it supplies the rest). Each is tracked here only as a dependency, not a commitment this project can unilaterally adjust. Sequence is expressed in terms of the Must item that first consumes the dependency, not in calendar time.

Shared-platform dependencies are surfaced at monthly steering. If any upstream owner declares **amber** or **red** on a component in Table 11.5, the affected downstream Must milestone is re-planned at the following steering.

## 11.7 v2.0 expansion trigger

The Could item C01 (REG-V2-EXPLORE) is unlocked only when all of the following are observed after two consecutive quarters of full rollout under v1.0:

- a second regulator framework (beyond IMDA MGF) is formally adopted as in-scope by the AI Governance Working Group;
- MGF v1.1 pillars (S03) are fully mapped and at *compliant* status;
- business-case refresh (M07) explicitly budgets the v2.0 programme.

If any of the three conditions fails, C01 remains in the Could column and is re-evaluated at the next business-case refresh.

Table 11.3: Regulations launch tiers. The exit gate defines the MoSCoW items that must be closed to exit the tier.

| **Tier** | **Scope** | **Exit gate (references MoSCoW IDs)** |
|---|---|---|
| Alpha | Engineering only, DEV + UAT, conformance fixtures only. | M01, M02, M04 closed; conformance tooling green on fixtures; reproducibility of audit outputs confirmed on two consecutive runs. |
| Beta | Audit + identity + monitoring live for one consumer project (TB) in UAT. | M03, M05 closed; parallel-run gate in the TB spec passes for one full cycle; no open SEV1/SEV2; audit coverage = 100 % of in-scope TB actions. |
| GA (limited) | Audit + identity + monitoring live for TB in production. | M06 closed; Beta gate sustained for two consecutive cycles; kill-switch green; REG-MGF-2.1.2-002 at *partial* (TB-only). |
| Full rollout | All three current consumer projects integrated (TB, AP, Simula). | S01 and S02 closed; REG-MGF-2.1.2-002 at *compliant*; two quarterly reviews sustained; M07 complete; business-case refresh complete. |

Table 11.4: Regulations kill-switch.

| Condition | Evaluation | Action |
|---|---|---|
| Audit-coverage gap (missed `reserve/complete` pair) detected in production | continuous | **Halt GA onboarding for all consumers**; incident-review before next integration. |
| Circuit breaker tripped with no successful graceful-degradation fallback for three consecutive cycles | per cycle | **Halt upstream LLM traffic**; fail over to cached/he path; remediate before resuming. |
| Identity-attribution reconciliation fails (principal of record cannot be resolved) for $> 0.01\,\%$ of actions | per cycle | **Pause affected consumer**; block further reserves until remediated. |
| Any IMDA MGF finding attributable to the Regulations platform (audit, identity, monitoring) | continuous | **Halt rollout**; remediate before next wave. |
| Any consumer project declares red on a Regulations-supplied component | continuous | **Pause** |

Table 11.5: Shared-platform dependencies.

| Component | Owner | First consumed by | Impact if late |
|---|---|---|---|
| Unified schema registry (`docs/schema/`) | Shared infra | M01 | Already `migration.status=complete`; no remaining risk. |
| vLLM hosting (`vllm-only` policy) | Shared platform | M04 | The graceful-degradation ladder (Section 8.5.4) is the mitigation; this project is the *owner* of that ladder for sibling consumers. |

## 11.8 Governance and business-case cadence

**Project governance.** The Regulations project reports monthly to a project steering (AI Governance Lead, Compliance Officer, Platform Engineer) and quarterly to the cross-project GFAIC, which it also convenes. Monthly steering accepts or rejects amber/red signals on Must items; GFAIC reviews compliance posture and cross-project dependencies (this project's consumer-integration Should items feed directly into sibling projects' Must items).

**Business-case refresh (M07).** The business case that authorises this project is refreshed on the following rhythm:

- **Mandatory refresh**: at the first cycle after GA, and annually thereafter (ticket REG-BC-REFRESH, priority M in Table 11.2). Refresh packet includes consumer integration load assumptions, unit economics and platform operating cost realised vs. forecast, and a deviation statement against the baseline.
- **Trigger-based refresh**: any of
  - consumer-project count in scope changes by $\geq 1$ (new consumer adopts, or one is retired);
  - per-action audit cost changes by $\geq 30\,\%$ (storage, indexing, identity-reconciliation throughput);
  - a new MGF pillar, a new regulator framework, or a material change to REG-MGF-2.1.2-002;
  - any kill-switch in Table 11.4 is tripped.
- **Owner**: AI Governance Lead. Approval routed through the project steering and filed at `docs/regulations/business-case/refresh-history.yaml`.

# Chapter 12

# Agent Implementation Instructions

## 12.1 Purpose and audience

This chapter is the self-contained, intent-based instruction set for implementing the *Regulations* domain inside the unified SAP OSS platform. Regulations plays two roles at once: it is a persona surface (Compliance Officer) in its own right, *and* it is the horizontal runtime wrapper that governs the three other domains (Arabic AP, TB, Simula) at every MCP call. This chapter is written for LLM-driven and human implementation agents who will extend existing services to realise both roles. It assumes only this PDF; every platform decision, integration contract, and governance obligation is restated here so that no companion document is required. Existing chapters 1–11 remain the normative source for the regulations domain itself; this chapter tells future agents *how* to build what those chapters describe, *where* to put it in the codebase, and *how* it plugs into the wider platform.

## 12.2 Rules of Engagement for AI Agents

To ensure safety and architectural integrity, all implementation agents must adhere to the following Prime Directives:

RULE-1: **Credential Protection.** Never log, print, or commit secrets. Use the `.secrets.example` pattern for all new integrations.

RULE-2: **Context Efficiency.** Prioritize surgical edits. Request only the minimum required file ranges to satisfy the task intent.

RULE-3: **Test-Driven Implementation.** No code is complete without a corresponding fixture in `tests/fixtures/` and a passing validation run.

RULE-4: **Self-Correction.** If a schema validation fails, the agent must diagnose the root cause (hallucination vs. schema drift) before re-attempting implementation.

RULE-5: **Safety Inheritance.** Domain-specific agent rules (`.clinerules`) may only *tighten* (e.g., lower timeout, stricter precision), never *loosen* regulatory constraints.

## 12.3 Agent Rule-Pack Architecture (.clinerules)

The platform mandates a three-tier inheritance hierarchy for agent behavior. Implementation agents must load the relevant rule pack before starting any task.

### 12.3.1 Inheritance Hierarchy

1. **Repository-level (`/.clinerules`):** The baseline for all engineering and documentation standards.
2. **Intelligence-level (`/src/intelligence/.clinerules`):** The regulatory envelope (MAS MGF) that wraps all domains.
3. **Domain-level (`/src/<domain>/.clinerules`):** Domain-specific constraints (e.g., Arabic character precision).

### 12.3.2 Safety-Critical Flags

The following flags in the rule packs are subject to RULE-5 (Inherit, Never Weaken):

Table 12.1: Safety-Critical Agent Flags.

| Flag | Parent Value (Regs) | Tightening Direction |
|---|---|---|
| `deny_by_default` | `true` | Invariant (Cannot be set to false). |
| `require_identity` | `true` | Invariant (Cannot be bypassed). |
| `max_retries` | `3` | May only *decrease*. |
| `timeout_ms` | `30000` | May only *decrease*. |

## 12.4 Platform Overview — One Codebase, One Instance

### 12.4.1 Integration Stance

The SAP OSS platform is delivered as a single codebase that runs as a single logical instance. All cross-cutting concerns are hosted by existing services.

**One Codebase, One Instance**

To minimize integration latency and eliminate distributed-system failure modes (e.g., partial partitions), we mandate a single logical instance. All cross-domain orchestration happens via the MCP Gateway layer.

### 12.4.2 Platform Integration Map

## 12.5 Transactional Sequence Diagrams

Implementation agents must adhere to the temporal logic defined below. Every cross-domain request MUST be brokered by the Regulations Wrapper.

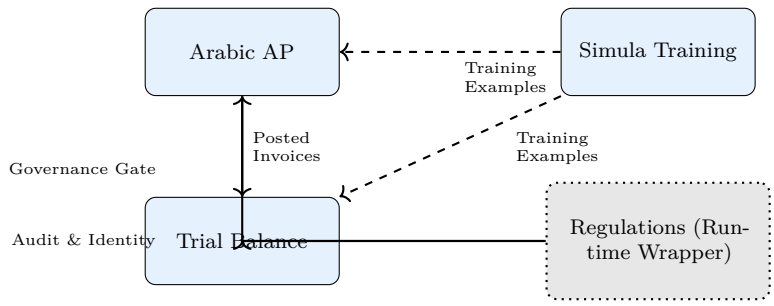


Figure 12.1: Cross-domain interaction and unified data flow map. Detailed governance logic is defined in **??**.

### 12.5.1 Universal Governance Handshake

This sequence applies to **all** platform service calls.

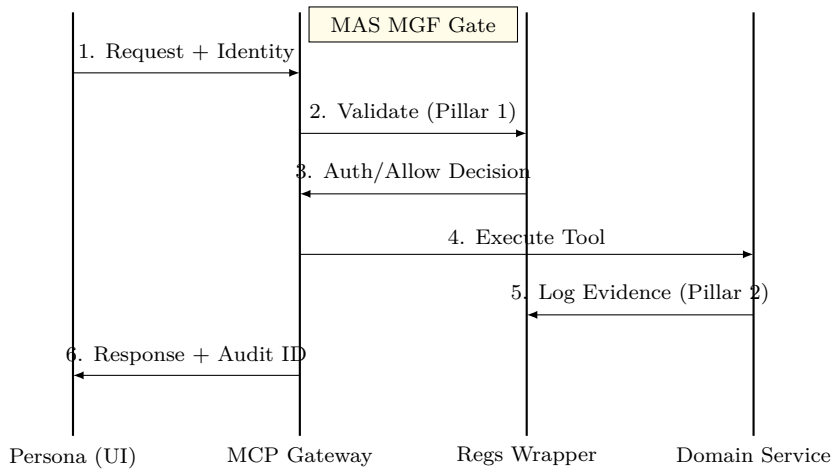


Figure 12.2: Standard platform handshake for governed tool execution. Audit schemas are detailed in **??**.

### 12.5.2 Arabic AP Golden Path

Temporal flow for invoice processing with Arabic context.

## 12.6 Compute and Resource Alignment

AI algorithms are calibrated for the reference infrastructure established in the domain chapters:

Table 12.2: Infrastructure Reference Profiles.

| **Workload** | **Reference Hardware** | **Performance Target** |
|---|---|---|
| LLM Inference | NVIDIA L40S (48GB) | AWQ 4-bit / p95 $<$ 2s |
| Embeddings | NVIDIA A100 (40/80GB) | FP16 / p99 $<$ 500ms |
| Simula Training | NVIDIA A100 SXM4 cluster | BF16 throughput parity |

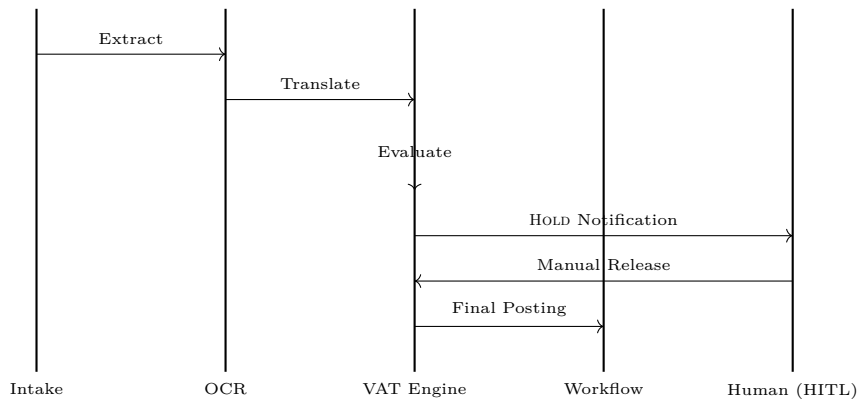


Figure 12.3: Arabic AP sequence: Scanned PDF to Release. VAT rules are specified in **??**.

## 12.7 Human-in-the-Loop (HITL) Critical Gates

The following "Four-Eyes" gates are mandatory. AI agents must implement the "Awaiting Sign-off" state for these actions:

- **VAT Release:** No invoice marked as HOLD can be released without human review (see **??**).
- **Anomaly Override:** Statistical outliers ($3\sigma$) cannot be dismissed by AI; only "Confirmed" or "Triaged" by human (see **??**).
- **Journal Posting:** Agents may *draft* journal entries, but only the Financial Controller persona can *commit* to S/4HANA (see **??**).

## 12.8 Registry Integrity & Live Status

Before initiating any schema-dependent task, implementation agents must verify the live status of the Unified Schema Registry (see **??**).

> **Registry First**
>
> If `docs/schema/registry.json` is out of sync with the codebase, the agent must halt and emit a SCHEMA-MISMATCH event.

## 12.9 Master Implementation Manifest (Union)

Implementation runs in three waves across the entire platform.

### Wave 1 — Foundations

- **Simula:** Implement HANA schema extraction pipeline (MCP discovery). See **??**.
- **Regulations:** Implement MAS MGF Pillar 1 logic (Tool allow-list). See **??**.
- **Arabic:** Implement Invoice Intake & OCR Gateway. See **??**.

- **Trial Balance:** Implement identity-attribution wiring. See **??**.

### Wave 2 — Core Domain Build-out

- **Trial Balance:** Implement Controls & Commentary Engine ($3\sigma$ anomaly logic). See **????**.
- **Arabic:** Implement multi-country VAT Engine (Saudi, UAE, Bahrain, Oman). See **????**.
- **Simula:** Implement Taxonomy Generation & Agentic Data Synthesis. See **????**.
- **Trial Balance:** Realise 28-state BPMN workflow (`tb-review-glo`). See **??**.

### Wave 3 — Hardening

- **Regulations:** Implement Per-Request Identity Attribution. See **??**.
- **Trial Balance:** Deliver Human Review Dashboard. See **??**.
- **Simula:** Deliver test fixtures and testing at 85% coverage. See **????**.
- **Regulations:** Wire Emergent-Capability monitoring. See **??**.

## 12.10 Governance Obligations (MAS MGF)

The regulations domain wraps all service calls at runtime. Agents must implement the following hooks in every domain:

- **Pillar 1:** Tool allow-list enforcement (deny-by-default).
- **Pillar 2:** Traceability via the identity attribution envelope.
- **Pillar 3:** Metric emission to the monitoring sink.

Detailed pillar requirements are defined in **??**.

## 12.11 FMEA: Failure Mode and Effects Analysis for AI

The following analysis identifies critical AI failure modes within the Regulations governance runtime and their deterministic mitigations.

Table 12.3: FMEA for Regulations AI Governance Pipeline.

| **Failure Mode** | **Symptom** | **Impact** | **Technical Mitigation** |
|---|---|---|---|
| Entitlement Drift | Persona allowed to call a non-allow-listed MCP tool. | Critical (Security) | Deny-by-default hard gate in MCP Gateway middleware. |
| Audit Omission | High-risk action emitted without Identity Envelope. | High (Compliance) | Mandatory envelope-check hook in PAL evidence-writer. |
| Monitoring Blindspot | Emergent capability detected but metrics not emitted. | Medium (Risk) | Periodic 'Heartbeat' probe to Monitoring Sink. |

## 12.12 This domain in context

Regulations is unique among the four domains: it is both a persona surface (Compliance Officer) and the horizontal runtime wrapper that governs Arabic, TB, and Simula at every MCP call. As a persona surface, its working view is the governance dashboard surfaced through the shared workspace shell and gated by regulations entitlements. As a runtime wrapper, it has no separate service of its own: the governance hooks are attached to the gateway and to ai-core-pal, and they are exercised on every call the platform handles. Its entry points are (a) any inbound MCP call into any domain, which must pass through the governance gate, and (b) the Compliance Officer's review queue. Its exit points are (a) governance decisions (allow, deny, degrade) returned to the gateway, (b) evidence records written to the shared governance store, and (c) conformance reports for auditors.

## 12.13 Schema contribution

Regulations owns the schemas catalogued in chapter 2 and listed in 2.1. The canonical set is the requirement record (the linkage schema between source rule, artefact, and evidence), the conformance-tool record, the enum vocabularies in 2.2, the capability-monitoring metrics record (chapter 8), and the identity and audit-event schemas in chapter 10. These schemas live under `docs/latex/specs/regulations/` as normative text and under `docs/schema/regulations/` as the JSON Schema artefacts consumed at runtime. Registration in the unified registry is additive: each schema is listed in `src/intelligence/ai-core-pal/data_products/registry.yaml` with its URI, owning domain (`regulations`), security class (`confidential`), and LLM routing (`vllm-only`), following the ODPS 4.1 pattern. Simula consumes these schemas directly as inputs to its extraction pipeline; it does not transform them. Regulations' `requirement.schema.json` is a distinct URI from TB's `requirement.schema.json`; the two files are not merged.

## 12.14 Persona and MCP allow-list

- **Persona id:** `compliance-officer`.
- **Roles:** requirement registration, governance-policy authoring, conformance-run triage, emergent-capability review, audit-trail inspection.
- **Entitlements:** read/write on the requirement record, conformance-tool record, capability-monitoring record, and the identity-and-audit schemas; read on every other domain's evidence output (needed to assemble conformance reports); override-write on governance-decision records (rare, audit-logged).
- **UI route:** `/governance/{tenant}/dashboard` on the shared workspace shell; surfaced only to callers whose entitlement set includes `compliance-officer`.
- **MCP tool allow-list:** `gov.requirement.register`, `gov.policy.publish`, `gov.conformance.run`, `gov.conformance.report`, `gov.monitor.query`, `gov.audit.query`, `gov.identity.attest`. All other MCP tools are denied by

default per IMDA MGF pillar 1.

## 12.15 Agent Rule-Pack Binding

Before initiating any task in this domain, the agent MUST load the following rule packs in order:

1. `.clinerules` (Global Standards)
2. `src/intelligence/.clinerules` (Regulations Envelope)

## 12.16 Implementation tasks for this domain

The tasks below are intent-based: each task states the outcome a future agent must achieve and the location in the existing codebase where the work belongs. None of them contains code; each task binds to the regulations chapters that define the required behaviour.

### Task R1 — MGF four-pillar runtime checks

**Intent.** Realise the four IMDA MGF pillars defined in chapter 3 as runtime checks that run on every inbound MCP call in every non-regulations domain.

> **Why Pillars over Flat Rules?**
>
> We adopted the IMDA MGF Pillar structure to ensure that governance is not just a 'checklist' but a runtime operational model that scales as new domains (e.g., ESG, Supply Chain) are onboarded.

**In-scope.** Pillar 1 (assess and bound) via allow-list enforcement at the gateway; pillars 2–4 via the evidence and disclosure hooks on ai-core-pal; pillar binding per §3.3 and 3.1. **Out-of-scope.** Re-implementing the MGF rule source; the source document is frozen and referenced by its sha in 7.1. **Paths to extend.** Governance hooks in the gateway; evidence writer in ai-core-pal. **Definition of done.** Every allow/deny decision references a pillar; denials are surfaced to the caller with a stable error code; evidence records for each pillar are written for every call.

**Autonomous Verification (Agent Logic):**

```
// Verify Task R1
{
  "test_fixture": "tests/fixtures/regulations/unauthorized_call.json",
  "expected_outcome": "http_status == 403",
  "validation_command": "make test-gov-runtime --pillar=1"
}
```

**Verification.** Running any non-regulations domain's golden sample produces a governance evidence record per pillar per call; `make spec-regulations` remains green.

### Task R2 — AI Agent Index mapping

**Intent.** Operationalise the 2025 AI Agent Index transparency requirements listed in chapter 4, especially 4.1. **In-scope.** Requirement registration per row; verification runners that check that each agent surface exposes the required transparency metadata (model, training set summary, known failure modes). **Out-of-scope.** Public disclosure portals. **Paths to extend.** Requirement record loader; governance dashboard transparency tab. **Definition of done.** Every row in 4.1 maps to a runnable verification; failures show up on the dashboard. **Verification.** Conformance runner report 100% of Agent Index requirements either passing or explicitly waived with attached evidence.

### Task R3 — Empirical evaluation hookup

**Intent.** Encode the requirements extracted from Ju & Aral (2026) in chapter 5 and 5.1 as runnable empirical checks against the platform. **In-scope.** Test harness that executes the Ju & Aral requirements on Arabic, TB, and Simula outputs; periodic run cadence. **Out-of-scope.** Adding new empirical studies. **Paths to extend.** Regulations test harness colocated with chapter 9 fixtures; scheduler hook. **Definition of done.** Every requirement in 5.1 has a callable check; the harness reports pass/fail per requirement on the most recent platform build. **Verification.** Harness run produces a per-requirement report; failures block the CI gate.

### Task R4 — Conformance tooling integration

**Intent.** Integrate AI Verify 2.0 and Moonshot CI/CD 1.1.0 per chapter 6 and the coverage map in 6.1. **In-scope.** AI Verify plug-ins listed in 6.1 and Moonshot benchmarks listed in 6.2; conformance-run CLI exposed to the Compliance Officer persona. **Out-of-scope.** Upstream changes to AI Verify or Moonshot. **Paths to extend.** Conformance harness invoked from the gateway; dashboard run-history view. **Definition of done.** Running the conformance suite against the platform produces a unified report covering every plug-in and every benchmark in the two tables. **Verification.** `gov.conformance.run` returns a report that validates against the conformance-tool schema.

### Task R5 — Capability monitoring

**Intent.** Deliver the capability monitoring specification in chapter 8, with the architecture in 8.1 and the metrics in 8.1. **In-scope.** Metrics collection at each of the collection points; storage in the monitoring sink; threshold-based alerts. **Out-of-scope.** Long-term analytics warehouse. **Paths to extend.** Monitoring hooks on ai-core-pal; alert router. **Definition of done.** Every metric in 8.1 is emitted in production; alerts fire on the thresholds the chapter

defines. **Verification.** Synthetic load test shows every required metric at the sink; alerting simulator triggers every configured alert.

### Task R6 — Identity attribution

**Intent.** Realise per-request identity attribution per chapter 10, including the agent identity, request identity, and audit-event schemas. **In-scope.** Identity envelope construction at the gateway; propagation through every domain service call; audit writer that emits every required event. **Out-of-scope.** Replacing the existing authentication backend. **Paths to extend.** Gateway identity middleware; audit writer in ai-core-pal. **Definition of done.** No MCP call leaves the gateway without a complete identity envelope; every required audit event is written with the schema's required fields. **Verification.** Audit query over a test run surfaces every event class; absent or partial envelopes fail the CI gate.

### Task R7 — Testing

**Intent.** Deliver the regulations testing specification in chapter 9 at the coverage thresholds in 9.1. **In-scope.** Regulation fixtures, conformance-tool fixtures, unit tests for requirement parsing, integration tests over the governance gate. **Out-of-scope.** Load benchmarking beyond what chapter 9 defines. **Paths to extend.** Test directory colocated with the regulations service code. **Definition of done.** Coverage meets 9.1; CI gates block regressions. **Verification.** `make spec-regulations` remains green; the regulations test entry points all run.

## 12.17 Governance obligations applied to this domain

Regulations is both the persona surface and the wrapper. The four obligations therefore apply to regulations in two ways: as the author of the obligations (canonical source in this spec's chapters) and as a subject of the obligations (like every other domain, its own MCP calls pass through the governance gate).

- **IMDA MGF four pillars** (chapter 3). The regulations service authors the pillar definitions; it also executes them against its own MCP tools. Pillar 1 enforcement applies the allow-list in §12.14, deny-by-default.
- **Identity attribution** (chapter 10). Regulations owns the identity and audit-event schemas; its own MCP calls carry the identity envelope like any other domain.
- **Emergent-capability monitoring** (chapter 8). Regulations authors the metrics model; governance decisions, conformance runs, and audit queries are themselves instrumented.
- **Conformance tooling** (chapter 6). Regulations integrates AI Verify 2.0 and Moonshot CI/CD 1.1.0 as runnable suites; the regulations service's own behaviour is included in every conformance run.

## 12.18 Cross-spec integration — full detail

### 12.18.1 With Arabic AP

Regulations wraps Arabic at runtime. Every Arabic MCP call is brokered through the regulations governance gate, which applies pillars 1–4 of IMDA MGF, attaches the identity envelope, and routes emergent-capability metrics to the monitoring store. Arabic exports the invoice record, VAT compliance checklist, and payment instruction schemas under `docs/schema/regulations/` and registers its MCP tools with the gateway; the governance gate consumes those registrations but adds no Arabic-specific code paths. Conformance tests defined here execute against the Arabic service as part of the platform CI gate.

### 12.18.2 With Trial Balance

Regulations wraps TB at runtime. Every TB MCP call is brokered through the regulations governance gate, which applies pillars 1–4 of IMDA MGF, attaches the identity envelope, and routes emergent-capability metrics to the monitoring store. TB exports the control record, decision-point record, workflow record, and its own `requirement.schema.json` under `docs/schema/regulations/`. Regulations' `requirement.schema.json` is a distinct URI from TB's; no schema merge takes place.

### 12.18.3 With Regulations

Regulations exports the requirement record, conformance-tool record, enum vocabularies, capability-monitoring metrics record, and the identity and audit-event schemas under `docs/schema/regulations/`; registers the MCP tools enumerated in §12.14; and applies the four governance hooks in §12.17 to every call it serves — including its own. External consumers (Arabic, TB, Simula) see regulations only through these contracts and through the wrapper; there are no private back-channels.

### 12.18.4 With Simula

Regulations wraps Simula at runtime in the same way it wraps Arabic and TB. Simula also consumes regulations' schemas (requirement, conformance-tool, monitoring metrics, identity and audit) as training input. The contract is schema-in, `TrainingExample`-out: Simula reads the regulations schema set; it produces `TrainingExample` records. Simula does not call regulations MCP tools at runtime and does not share state with the governance dashboard. No cross-domain operational envelope is introduced.

## 12.19 Wave assignment and sequencing

Implementation runs in three waves.

- **Wave 1 — foundations.** Identity schema (chapter 10) frozen; gateway identity envelope; persona-gated surface for the Compliance Officer; unified schema registry

entries for regulations schemas.

- **Wave 2 — governance runtime.** Tasks R1, R4, R5, R6 above. Prerequisites: Wave 1 complete; Simula Wave 1 extraction pipeline available for monitoring metrics ingestion.
- **Wave 3 — evaluation and hardening.** Tasks R2, R3, R7. Prerequisites: Wave 2 complete; monitoring sink live; all non-regulations domains exposing their conformance entry points.

Regulations is the critical-path domain for Wave 1 (identity) and Wave 2 (governance runtime). Other domains block on regulations' Wave 1 completion.

## 12.20 Acceptance criteria (platform-wide)

When all four chapter 12 instruction sets are implemented, the platform satisfies the following invariants:

1. **One instance.** A single codebase runs a single logical instance; no domain ships as a separate product.
2. **Persona-gated surfaces.** The four personas each reach a tailored UI on the shared shell; no route, MCP tool, or data product is visible outside its persona's entitlement set.
3. **Unified registry.** Every schema in use is listed in the ODPS 4.1 registry under its owning domain; validators resolve `$ref`s through the unified registry only.
4. **No schema merges.** The two `requirement.schema.json` files in TB and regulations remain distinct; no cross-domain composite schemas are produced.
5. **No cross-domain operational envelope.** Runtime orchestration stays within each domain's service; only Simula consumes other domains' outputs, and only as training input.
6. **Governance wraps all three non-regulations domains.** Arabic, TB, and Simula service calls pass through the regulations runtime wrapper without exception.

## 12.21 References

Within-spec references: chapter 1 (overview), chapter 2 (data schema), chapter 3 (IMDA MGF framework), chapter 4 (AI Agent Index), chapter 5 (Ju & Aral empirical evidence), chapter 6 (conformance tooling), chapter 7 (references and traceability), chapter 8 (capability monitoring), chapter 9 (testing specification), chapter 10 (identity attribution), chapter 11 (project roadmap).

Peer chapter 12 instruction sets for the other three domains (each is self-contained; consult directly when operating in that domain):

- Arabic AP: `docs/latex/specs/arabic/chapters/12-implementation-instructions.tex`.

- Trial Balance: `docs/latex/specs/tb/chapters/12-implementation-instructions.tex`.
- Simula: `docs/latex/specs/simula/chapters/12-implementation-instructions.tex`.

Shared platform artefacts:

- ODPS 4.1 registry: `src/intelligence/ai-core-pal/data_products/registry.yaml`.
- Schema pipeline and resolver: `src/intelligence/schema_pipeline/`.
- Training pipeline (Simula): `src/training/pipeline/`.
- Shared workspace shell: `generativeUI/training-webcomponents-ngx`.

# Chapter 13

# Global Wave-Dependency Manifest

## 13.1 Critical Path Visualization (Waves 1–3)

The following chart defines the normative sequencing for implementation agents. A task in a later wave cannot be marked COMPLETE until its prerequisites in an earlier wave are validated.

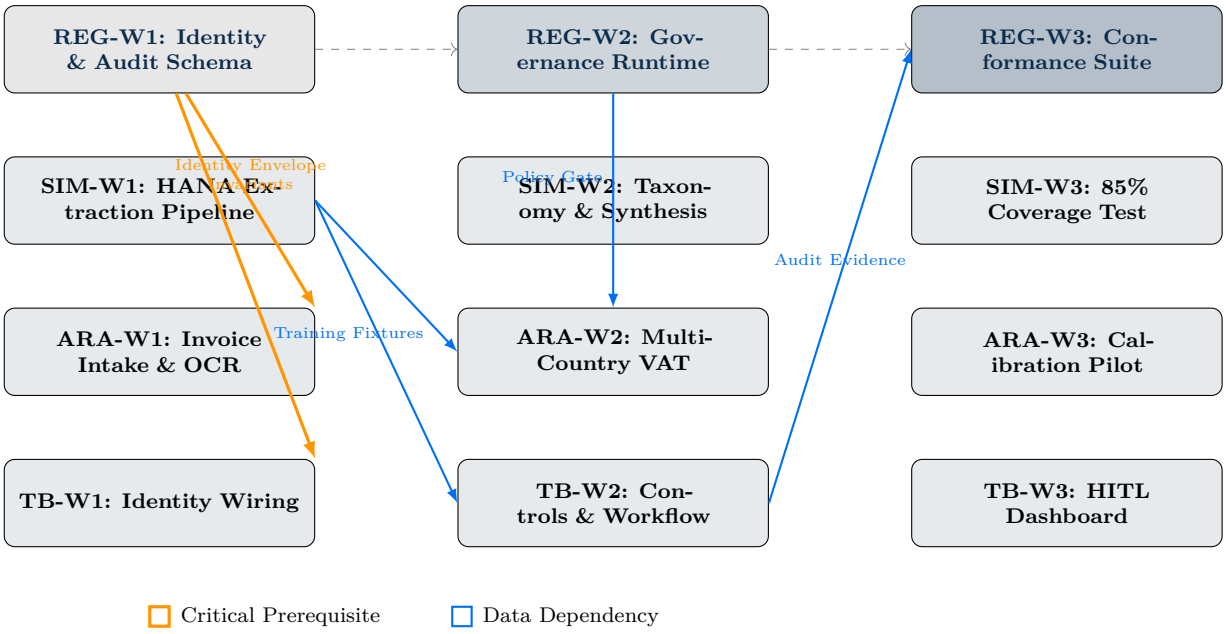


Figure 13.1: Cross-specification dependency graph: The blueprint for platform rollout.

Table 13.1: Normative Cross-Domain Prerequisites.

| Target Task | Prerequisite | Ref ID | Rationale |
|---|---|---|---|
| **ARA-W1** | REG-W1 (Identity) | `REG-IDENTITY-01` | Invoice intake must be audit-traced from Day 1. |
| **TB-W2** | SIM-W1 (Extraction) | `SIM-EXTRACT-01` | Anomaly detection requires HANA schema metadata. |
| **REG-W2** | SIM-W1 (Registry) | `SIM-REGISTRY-01` | Governance monitoring relies on the schema registry. |
| **ARA-W2** | REG-W2 (Pillar 1) | `REG-MGF-P1` | VAT release is an HITL gate governed by MAS Pillar 1. |
| **TB-W3** | ARA-W2 (Posted Invoices) | `ARA-AP-POST` | Review dashboard depends on posted AP data from Arabic domain. |

## 13.2 Inter-Domain Prerequisite Matrix

**Dependency Enforcement**

Implementation agents are strictly forbidden from "Fast-Forwarding" into Wave 2 if Wave 1 critical prerequisites (marked in **Gold**) are in a GAP or PARTIAL state.